CHEMISTRY GRADE 10: THE ATOM

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.2 (9 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Chemistry
Strand	Strand 1.0: Inorganic Chemistry
Sub-Strand	Sub-Strand 1.2: The Atom
Total Duration	9 lessons × 40 minutes = 360 minutes total
Sub-Strand Content	– Structure of the atom: energy levels and orbitals – Atomic number, mass number, and number of electrons – Historical atomic models: Dalton, Rutherford, and Bohr models – Isotopes and relative atomic mass – Calculation of relative atomic mass from isotopic abundances – Relationship between energy levels and orbitals – Order of filling electrons in orbitals (Aufbau principle, Pauli exclusion, Hund's rule) – s and p orbitals for the first 20 elements – Electron arrangement notation using s and p orbitals – Rutherford Gold Foil experiment (simulation and discussion)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe the structure of the atom, b) determine the relative atomic mass of elements, c) write the electron arrangement of elements using s and p notation, d) develop interest in the study of structure of the atom.
Core Competencies	– Digital Literacy: The learner continuously learns to use digital platforms while watching animations on atomic models, isotopes and electron arrangement of atoms — embedded when students watch the Rutherford Gold Foil simulation and animations of orbital filling sequences on the ARES platform. – Creativity and Imagination: The learner experiments with ideas while modelling the structure of the atom using locally available materials — embedded when students construct physical atom models using locally sourced materials such as maize seeds, bottle tops, and wire found in Kenyan markets. – Learning to Learn: The learner reflects on their own work as they practise illustration of electron arrangements for the first 20 elements using s and p orbitals — embedded through self-assessment worksheets and peer review of orbital notation exercises across lessons 7–9.
Core Values	– Social Justice: The learner advocates harmonious relationship in groups while discussing the meaning of isotopes and relative atomic mass — upheld through structured collaborative group discussions where every student's contribution is valued regardless of background.

	– Integrity: The learner shows self-discipline by appropriately using digital devices to watch animations, simulations or videos on electron arrangement of the atom — reinforced through clear device-use agreements and teacher monitoring during simulation lessons on the ARES platform.
Science & Engineering Practices	– SEP 2 – Developing and Using Models: Students construct and revise physical and diagrammatic models of the atom (Dalton, Rutherford, Bohr) to explain how atomic structure has been refined over time, directly mirroring how scientists build and improve explanatory models. – SEP 4 – Analysing and Interpreting Data: Students analyse isotopic abundance data for elements such as chlorine and carbon to mathematically calculate relative atomic mass, interpreting patterns between isotope ratios and weighted averages. – SEP 6 – Constructing Explanations: Students use evidence from the Rutherford Gold Foil simulation to construct a scientific explanation for why most alpha particles passed through the gold foil while a few were deflected at large angles, leading to the nuclear model. – SEP 8 – Obtaining, Evaluating, and Communicating Information: Students watch digital simulations, read KICD-aligned texts, and communicate their understanding of electron arrangement through s and p notation diagrams shared with peers during DQB updates.
Pertinent & Contemporary Issues (PCIs)	– Socio-Economic and Environmental Issues (Environmental Conservation): The learner responsibly uses locally available materials to model the structure of the atom — students select and repurpose everyday Kenyan materials (bottle tops, dried beans, maize cobs) reducing waste while building atomic models. – Life Skills (Self-Esteem): The learner gains confidence while presenting findings on the meaning of isotopes and relative atomic mass — achieved through structured gallery walks and short oral presentations where students share calculated relative atomic masses to the class. – Socio-Economic and Environmental Issues (Global Citizenship): The learner brainstorms with peers on the contributions of scientists in the development of atomic theory/models — students explore the global timeline of atomic theory from Dalton to Bohr, discussing how international scientific collaboration shapes shared knowledge.
Career Connections	– Nuclear Medicine Technologist: Understanding atomic structure and isotopes is foundational for careers using radioactive isotopes in medical imaging and treatment at facilities such as Kenyatta National Hospital. – Chemical Engineer: Electron arrangement and atomic theory underpin the design of industrial chemical processes in Kenya's manufacturing sector, including fertiliser production at Eldoret-based agro-industries. – Materials Scientist / Nanotechnologist: Knowledge of atomic structure and orbital theory is essential for developing new materials, relevant to Kenya's growing technology and innovation hubs such as Konza Technopolis. – Secondary School Chemistry Teacher: Mastery of atomic models and electron configuration prepares learners to communicate these foundational concepts to future Kenyan students under the CBC framework. – Pharmacist / Pharmaceutical Researcher: Understanding isotopes and atomic mass supports work in drug formulation and quality control in Kenya's expanding pharmaceutical manufacturing industry.
Focus for Lessons	This unit anchors the study of atomic structure within the puzzling observation of coloured flames produced when common Kenyan substances — like table salt (sodium chloride) or potassium-rich wood ash — are burned, a sight familiar from cooking fires across Nairobi to the Rift Valley. Students progress from reviewing Dalton's solid-sphere model through Rutherford's nuclear model and Bohr's energy-level model, building toward a modern understanding of s and p orbitals and electron configuration for the first 20 elements. Throughout the nine lessons, students use physical models built from local materials, digital simulations, and collaborative sensemaking to connect the invisible world of electrons to observable phenomena they encounter in everyday Kenyan life.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside an atom?

	KICD KEY INQUIRY QUESTION: 1. How are electrons arranged in an atom?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Colours of Fire at a Kenyan Cooking Hearth At a traditional Kenyan homestead in the Rift Valley, a family burns dry firewood to cook ugali. When the cook accidentally spills a pinch of salt (sodium chloride) into the flame, the fire briefly glows a brilliant yellow-orange. A neighbour adds potassium-rich banana-peel ash to their own fire and sees a soft violet flash. Later, at a school chemistry laboratory in Kisumu, students observe the same mystery: different substances — copper wire, table salt, steel wool — each produce a distinct, repeatable colour when placed in a Bunsen burner flame. WHY IT IS PUZZLING: Students naturally ask: Why does fire change colour depending on what substance is inside it? All substances seem to be made of atoms, and fire is just heat — so why would the type of atom matter? Why does sodium always give yellow and potassium always give violet? The puzzle is deepened by the fact that the colours disappear when the substance is removed and reappear when it is added back. Students cannot yet explain this with prior knowledge. This phenomenon is puzzling, observable, and culturally grounded — it is something Kenyan learners have seen at home, at celebrations like nyama choma evenings, or in school labs, making it immediately relevant. The resolution requires understanding that electrons within atoms occupy specific energy levels, can absorb energy from the flame, jump to higher levels, and release that exact energy as visible light of a specific colour when they fall back — meaning the colour is a fingerprint of atomic electron structure. This drives the entire unit's investigation into how electrons are arranged inside atoms.
Storyline Thread	Lesson 1 — Noticing and Wondering (Anchoring Phenomenon Introduction): Students observe a teacher demonstration of flame tests using sodium chloride, potassium permanganate, and copper sulphate solutions in a Bunsen burner. Students record observations, make initial predictions about why different colours appear, and post their "wonder questions" on the Driving Question Board (DQB). The lesson ends with students unable to fully explain the phenomenon — creating productive intellectual tension that motivates the unit. Lesson 2 — What Is an Atom? Reviewing Atomic Structure (Dalton & Rutherford Models): Students review atomic number, mass number, and the number of protons, neutrons, and electrons through a peer discussion activity using element cards for the first 20 elements. They then watch the Rutherford Gold Foil simulation on the ARES platform, construct explanations for the surprising deflection results, and build a class Rutherford model using locally available materials (bottle tops for nucleus, maize seeds for electrons). Students connect back to the phenomenon: Rutherford showed atoms have a tiny, dense nucleus — but where are the electrons, and how does that relate to flame colours? Lesson 3 — Bohr's Model and Energy Levels: Students are introduced to Bohr's planetary model, learning that electrons occupy specific energy levels (shells). Using diagrams, they draw Bohr models for the first 10 elements. A Think-Pair-Share discussion asks: "If electrons can only be at certain energy levels, what happens when they gain energy from a flame?" Students update the DQB — connecting Bohr's model to the anchoring phenomenon: electrons jump to higher energy levels in a flame and release light when they fall back. Lesson 4 — Isotopes and Relative Atomic Mass (Part 1 — Meaning and Concept): Students brainstorm the meaning of "isotopes" using carbon-14 dating of Kenyan archaeological sites as a context. Through a structured group discussion, students distinguish isotopes of hydrogen (protium, deuterium, tritium) and chlorine (Cl-35, Cl-37) using atomic number and mass number. They connect to the DQB: isotopes have the same electron arrangement (same atomic number) — reinforcing that electron

	arrangement, not mass, determines chemical behaviour and flame colour. Lesson 5 — Isotopes and Relative Atomic Mass (Part 2 — Calculation): Students receive isotopic abundance data for chlorine (Cl-35: 75%, Cl-37: 25%) and carbon (C-12: 98.9%, C-13: 1.1%) and calculate relative atomic mass using the weighted average formula. A gallery walk allows groups to present their calculations. Life-skills PCI is embedded: students gain confidence presenting their computed values. DQB is updated to reflect that atomic mass is an average — reinforcing that the identity of an element (and its flame colour) is determined by its atomic number, not its mass. Lesson 6 — From Energy Levels to Orbitals (s and p Orbitals Introduction): Students transition from Bohr's shells to the modern orbital model. Using animations on the ARES platform (Digital Literacy), students observe the shapes of s and p orbitals. A card-sort activity has students match orbital names (1s, 2s, 2p, 3s, 3p) to their energy levels and electron capacities. Students revisit the DQB driving question: flame colours arise because electrons in specific orbitals absorb and release precise amounts of energy — the colour is a signature of the orbital structure of that atom. Lesson 7 — Filling the Orbitals: Aufbau, Pauli, and Hund's Rules: Students carry out a structured activity using "electron seat" diagrams (boxes representing orbitals) to discover the rules governing how electrons fill orbitals. Through guided inquiry, they infer the Aufbau principle (fill lowest energy first), Pauli exclusion principle (max 2 electrons per orbital, opposite spins), and Hund's rule (one electron per orbital before pairing). Students practise writing orbital configurations for elements 1–10. Connection to phenomenon: the specific orbital an electron occupies determines exactly how much energy it releases as light — explaining why sodium's 3s→2p transition always gives the same yellow colour. Lesson 8 — Electron Configuration for Elements 1–20 (s and p Notation): Students systematically write the full s and p electron configurations for all first 20 elements (H through Ca), working in pairs. A peer-teaching exercise has students explain configurations to each other (Learning to Learn competency). They identify patterns: elements in the same group (e.g., Li, Na, K) have the same outermost orbital type — connecting to the anchoring phenomenon via Think-Pair-Share: "Why do Group 1 metals like sodium and potassium produce similar but distinct flame colours?" Students update the DQB with a key insight: similar outer electron configurations → similar behaviour; slight differences in energy levels → different colours. Lesson 9 — Model Building, DQB Resolution, and Unit Consolidation: Students construct a final cumulative model — a poster or digital diagram — showing the complete journey from Dalton's solid sphere to Bohr's shells to the modern s/p orbital model, annotated with explanations of isotopes, relative atomic mass, and electron configuration for sodium and potassium. Groups present their models and use them to fully answer the driving question: the different flame colours of sodium (yellow) and potassium (violet) arise because their outermost electrons occupy orbitals at different energy levels, releasing different amounts of energy as light of different colours when excited by the flame. The DQB is formally closed as students evaluate which original wonder questions have been answered and which remain open for future strands (e.g., the Periodic Table in Sub-Strand 1.3).
Supporting Phenomena	– Neon Signs Along Nairobi's Tom Mboya Street: Students observe that neon signs glow in distinct colours (red-orange for neon, blue for argon) — puzzling because the gases themselves are colourless, yet when electricity passes through them they emit specific colours, hinting that electrons in different atoms have different allowed energy states. – Carbon-14 Dating of Ancient Kenyan Artefacts: Archaeologists at sites like Olorgesailie use carbon-14 (an isotope of carbon-12) to date ancient tools — students find it puzzling that two atoms of the same element (carbon) can have different masses and different stabilities, raising the question of what makes isotopes different at the atomic level. – The Rutherford Gold Foil Surprise: When alpha particles are fired at a thin sheet of gold foil, most pass straight through but a tiny fraction bounce almost directly backward — this is deeply counterintuitive if atoms were solid spheres (Dalton model), and it anchors the lesson on Rutherford's nuclear model and the mostly-empty nature of the atom.

	– X-Ray Machines at Kenyan Hospitals (e.g., Aga Khan Hospital, Nairobi): X-rays pass through soft tissue but are blocked by bone — students are puzzled by how radiation interacts selectively with matter, connecting to the idea that electrons in different atoms are arranged differently and interact with energy in distinct ways.

LESSON 1 (40 minutes): The Mystery of Coloured Flames: Noticing and Wondering	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the unit by anchoring student curiosity in an observable, culturally familiar phenomenon — coloured flames at a Kenyan cooking hearth — and to generate the productive intellectual tension that will drive the entire sub-strand investigation into electron arrangement inside atoms.
Knowledge	Students will know that different substances produce characteristic and repeatable colours when burned in a flame, and that this phenomenon is linked to the identity of the atoms present in each substance.
Skills	Students will be able to make careful observations, record data systematically, generate testable wonder questions, and post those questions onto a Driving Question Board.
Attitudes	Students will demonstrate curiosity about natural phenomena, openness to uncertainty, and willingness to share initial ideas even before they have full scientific explanations.
Key Inquiry Question	How are electrons arranged in an atom? — This lesson opens the inquiry by establishing that different atoms behave differently in a flame, hinting that something inside each atom — specifically the arrangement of electrons — must be responsible.
Purpose in Storyline	This is the opening anchor lesson. Its sole purpose is to make the phenomenon visible, puzzling, and personally relevant so that every subsequent lesson (atomic models, Bohr shells, orbitals, electron configurations) feels like a genuine attempt to answer questions students themselves have asked. The DQB created here will be updated at the end of every lesson through Lesson 9.
Safety Notes	Wear safety goggles throughout the demonstration. Teacher only handles the Bunsen burner, nichrome wire loops, and chemical solutions — students observe from their seats at a minimum distance of 1.5 metres. Solutions used: saturated sodium chloride, 0.1 M potassium permanganate, and 0.1 M copper sulphate — all low-hazard at these concentrations but must not be ingested. Copper sulphate is a mild irritant; teacher wears disposable gloves. Ensure the room is ventilated. A damp cloth and sand tray must be on the demonstration bench. Long hair must be tied back. No student approaches the flame without explicit teacher instruction.

B. LESSON OVERVIEW

In this 40-minute opening lesson, the teacher frames a culturally grounded story about a Kenyan cooking hearth in the Rift Valley where spilled salt turns a fire brilliant yellow-orange, and banana-peel ash creates a soft violet flash. Students first predict what they think will happen when different substances enter a Bunsen burner flame. The teacher then performs a live flame test demonstration using sodium chloride, potassium permanganate, and copper sulphate, while students record observations silently. A structured whole-class discussion surfaces the puzzling contradiction — all substances are made of atoms, fire is just heat, so why does the type of atom change the colour? Students cannot resolve this yet. They then create the class Driving Question Board by posting sticky-note wonder questions. Finally, students sketch a very rough initial model of what they think might be happening inside an atom during a flame test. No answers are given — the lesson ends in productive intellectual tension.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** [Second Law of Thermodynamics and entropy] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [topical-test-form-1-chemistry-introduction-to-chemistry-answers] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Students listen as the teacher narrates the Kenyan cooking hearth story (Rift Valley family cooking ugali; spilled salt makes the fire flash yellow-orange; a neighbour sees violet from banana-peel ash). The teacher shows three labelled sample jars: sodium chloride (table salt), potassium permanganate solution, and copper sulphate solution. Each student individually writes in their exercise book: (1) What colour, if any, do you predict each substance will produce in the flame? (2) Why do you think that? Students have had no prior instruction on this topic, so predictions will be intuitive — for example, that all flames are orange, or that coloured solutions make coloured flames. After 2 minutes of individual writing, students share predictions briefly with their desk partner.	Teacher narrates the hearth story with cultural warmth: 'Picture a homestead near Nakuru — a mother is cooking ugali on a wood fire. She reaches for salt and a pinch falls into the flame. The fire suddenly flashes a bright yellow-orange. Her neighbour, using banana-peel ash from her potassium-rich soil, sees a violet glow. You may have seen something like this at home or at a nyama choma evening. Today we are going to recreate that mystery here in our laboratory.' Teacher holds up each jar in turn: 'Before I do anything, I want you to predict — in your book, right now — what colour, if any, you think will appear when I put each of these into the flame. There is no wrong answer. I want your honest thinking.' Teacher allows 2 minutes of silent individual writing. WAIT TIME: Do not speak or prompt during these 2 minutes — circulate silently and note the variety of predictions being written. After 2 minutes: 'Turn to your partner. You have 60 seconds. Share your predictions — do you agree or disagree?' After partner talk: 'Keep your predictions in your book. We will come back to check them.' Teacher does NOT reveal any answers at this stage.	Eliciting prior knowledge and activating intuition: by asking students to predict before observing, the teacher surfaces existing mental models (or lack thereof) about flame colour. This creates a cognitive anchor — students will compare what they predicted against what they actually see, amplifying the puzzling nature of the phenomenon. The cultural story primes students to connect the laboratory observation to lived experience, which research shows increases engagement and retention.	Teacher circulates during the 2 minutes of individual writing and makes mental notes of the spread of predictions. Key diagnostic information: Do students think all flames will be the same colour? Do any students already mention atoms or electrons? Do students link colour to the colour of the solution (copper sulphate is blue — will they predict a blue flame)? This informs how the teacher frames the post-observation discussion. No marks are given; this is purely diagnostic.
Observe Phase 📹 **VIDEO:** [Second Law of Thermodynamics and entropy] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Students observe the teacher demonstration in silence with their notebooks open. The teacher cleans a nichrome wire loop in hydrochloric acid, then dips it into sodium chloride solution and holds it in the Bunsen burner flame. A vivid yellow-orange flame appears. Students record in their books: the colour they see, how intense it is, and how long it lasts. The teacher repeats with potassium	Teacher adopts a calm, deliberate pace during the demonstration to allow careful observation: 'I am going to do this slowly. Your only job is to watch and write. Do not talk — just observe.' Teacher holds the nichrome loop in the sodium chloride solution visibly before inserting into the flame: 'Watch carefully. I am putting sodium chloride — ordinary table salt — into the flame now.' Inserts	Direct observation followed by immediate predict-observe-compare: the structured two-column annotation ensures students confront the gap between their intuitions and the data. The teacher deliberately withholds explanation during this phase — explanation comes later — so that the observation itself remains the focus. The repetition of the sodium chloride test reinforces the idea of	Teacher scans the two-column tables during the 2-minute annotation period. Key checks: Are students recording specific colours (yellow, violet, blue-green) or vague terms (bright, shiny)? Are students noting that the colour depends on the substance? Are any students noting that the colour of the solution (blue copper sulphate) does not match the flame colour (blue-green)? The

📖 **PDF:** topical-test-form-1-chemistry-introduction-to-chemistry-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	permanganate solution — a soft violet-lilac colour appears. Students record again. Finally, copper sulphate solution produces a blue-green flame. Students record. The teacher then repeats the sodium chloride test a second time so students can confirm the observation is repeatable. After all three tests, students have 2 minutes to review their predictions and annotate: 'I predicted... I actually observed...' in a simple two-column table.	loop. WAIT TIME: Allow 10 full seconds of silence after the yellow-orange colour appears before saying anything — let students look. Then quietly: 'Write exactly what you see. Colour, brightness, how long it lasts.' Teacher cleans the wire in dilute HCl between tests, explaining: 'I clean the wire each time so the previous substance does not contaminate the next test.' For potassium: 'This is potassium permanganate. Same procedure.' WAIT TIME: Again, 10 seconds of silence after the violet colour appears. For copper sulphate: same procedure and same 10-second silence. After all three: 'Now go back to your predictions. Draw a line down the middle of the next page. On the left write what you predicted. On the right write what you observed. You have 2 minutes.' After the 2 minutes: 'What do you notice about whether your predictions matched reality?' Invite 3-4 student responses. Teacher does NOT explain why — responses are acknowledged with neutral phrases such as 'interesting — note that down' or 'that is a good observation — keep it.'	repeatability, which is a key puzzling feature: this is not random, it happens every time with sodium.	last point is a particularly productive puzzler that can be highlighted in the next phase.
Explain Phase 📹 **VIDEO:** Second Law of Thermodynamics and entropy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** topical-test-form-1-chemistry-	Students engage in a whole-class structured discussion aimed at surfacing the puzzle — not resolving it. The teacher poses three sequential questions orally, giving students 1 minute of individual thinking time before each whole-class share. Question 1: 'What pattern do you notice in what we just observed?' Question 2: 'All three substances are made of atoms. Fire is just heat. So why would the type of atom change the	Teacher writes Question 1 on the board: 'What pattern do you notice?' WAIT TIME: 1 full minute of individual silent thinking — teacher watches the clock visibly so students understand the wait is intentional. Then: 'Share with the person next to you — 30 seconds each.' Then whole class: 'What patterns did you and your partner notice?' Teacher maps responses on the board (e.g., different substance — different colour;	Structured sequential questioning with mapped responses: the teacher uses a public concept web on the board to make student thinking visible and treat it as scientific data worth examining. The three questions are carefully sequenced — from observational pattern to conceptual puzzle to implication — following a gradual increase in cognitive demand. The teacher explicitly names and validates intellectual tension,	Teacher listens carefully during whole-class share for two things: (1) whether any student mentions electrons, energy levels, or atomic structure — if so, note the name and return to this student in Lesson 3; (2) whether students are forming productive questions versus superficial ones (e.g., 'it is because the chemicals are different colours' versus 'maybe the heat does something to the atoms themselves'). Teacher

introduction-to-chemistry-answers Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	colour of the flame?' Question 3: 'Sodium always gives yellow. Potassium always gives violet. What does that tell you about the relationship between the substance and the colour?' Students jot notes after each discussion. The teacher maps student ideas on the board in a concept web, using students own words without correcting or completing them. The lesson deliberately ends with the class unable to fully explain the phenomenon — the teacher names this intellectual tension explicitly and validates it as the starting point for the entire unit.	colour is repeatable; the colours are very pure and specific). Teacher does NOT evaluate responses — says: 'I am just collecting your observations right now.' Teacher then writes Question 2: 'All substances are atoms. Fire is heat. Why does the type of atom matter?' WAIT TIME: 1 minute silent individual thinking. Then: 'Who would like to share their thinking — even if you are not sure?' Collect 4-5 responses. Teacher maps these on the board in a separate column labelled 'Our ideas so far.' Teacher explicitly says: 'Notice that none of us can fully explain this yet — and that is exactly where we should be at the start of a new topic. The fact that we cannot explain it yet is not a problem. It is an invitation.' Teacher then poses Question 3: 'Sodium always gives yellow. Potassium always gives violet. What does that tell you?' WAIT TIME: 45 seconds. Collect responses. Teacher maps them. Finally: 'I want to tell you something. The answer to this puzzle is hidden inside every atom in this room — including the atoms that make up your own body. By the end of this unit, you will be able to explain exactly why sodium gives yellow and potassium gives violet, using the precise language of atomic structure. But for today, your job is simply to sit with the question.'	which is a key pedagogy for driving sustained inquiry. By not providing answers, the teacher preserves the motivational force of the puzzle.	makes a mental note of the most scientifically generative student ideas to use as anchors in later lessons. No formal assessment — this is a diagnostic listening opportunity.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Second Law of Thermodynamics	Each student receives two sticky notes (or small pieces of paper if sticky notes are unavailable). On the first, they write one question they genuinely want answered about what they just observed — beginning with Why, How, or	Teacher distributes sticky notes (or paper slips): 'You each get two slips. On the first, write one genuine question that this demonstration has raised for you — a question you actually want answered. Start it with Why, How,	The Driving Question Board is the central sense-making infrastructure for the entire unit. By having students generate the questions themselves, the teacher ensures that the inquiry feels owned by the class rather than	The sticky notes are a rich formative data source. Teacher photographs or copies the board after the lesson. Key questions to analyse: Are student questions surface-level (what colour does X make?) or mechanistic (why does

[and entropy](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **PDF:** [topical-test-form-1-chemistry-introduction-to-chemistry-answers](#) Source: Kenya Curriculum Tools 2025 🔍 [Search ARES for similar readings](#)	What. On the second, they write one thing they think they already know that might be relevant. Students walk to the board (or the teacher collects and posts them) and attach their notes to a large sheet of paper or section of the classroom wall labelled 'OUR DRIVING QUESTION BOARD — Sub-Strand 1.2 The Atom.' The teacher reads several questions aloud, groups similar ones under emerging category headings (e.g., 'Questions about what happens inside the atom,' 'Questions about why colours are different,' 'Questions about where the colour energy comes from'), and writes the unit Driving Question at the top of the board in large letters: 'Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside an atom?' The board remains visible for the entire unit.	or What. On the second, write one thing you think you already know that might help us begin to answer the question. You have 2 minutes.' WAIT TIME: 2 full minutes of silent individual writing. Teacher then: 'Bring your slips to the board — or pass them forward.' As slips are posted, teacher reads several aloud with enthusiasm: 'Listen to this question from one of your classmates: Why does sodium always give the same colour every single time? That is a brilliant scientific question — because it tells us the colour must be a property of sodium itself, not just of the flame.' Teacher groups questions visibly and labels categories. Teacher points to the Driving Question at the top: 'This is the question that will guide everything we do in this sub-strand. Every lesson from now on, we will update this board as we learn more. Some of your questions will be answered in Lesson 3, some in Lesson 6, some not until Lesson 9. By the time we reach the end, this board will be a map of our thinking journey.' Teacher explicitly tells students: 'The DQB belongs to you. I will not remove any question without the class agreeing that it has been answered.'	imposed by a curriculum. The public grouping of questions makes the intellectual structure of the sub-strand visible. The two-slip structure (question plus prior knowledge) prevents the board from being purely a list of unknowns — it also surfaces the knowledge resources students already bring, which the teacher can build on in subsequent lessons.	the type of atom determine the colour)? Do any students mention electrons, protons, or energy — showing partial prior knowledge? Are there misconceptions visible (e.g., the colour comes from the colour of the solution)? This analysis directly informs the pacing and emphasis of Lessons 2 and 3.
Model Building Phase 📹 **VIDEO:** [Second Law of Thermodynamics and entropy](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#)	In the final 5 minutes, each student draws a very rough initial model in their exercise book. The prompt on the board reads: 'Draw what you think is happening inside a sodium atom when it is placed in a flame. Use arrows, labels, or whatever makes sense to you. There is no right answer yet — this is your starting idea.' Students draw independently and silently. The	Teacher writes the model-building prompt on the board and says: 'Before we finish today, I want a snapshot of your thinking right now — before you have learned anything new. Draw what you think is happening inside a sodium atom in the flame. It might be completely wrong. That is fine — in fact, that is the point. In Lesson 9, you will look back at this drawing and see	The initial model serves as a baseline representation of student understanding at the start of the unit. By explicitly telling students the model will be revised, the teacher establishes a growth mindset frame for the entire sub-strand — models in science are not final truths but evolving explanations. The act of drawing also surfaces visual	Teacher circulates during drawing and notes the range of model types: students who draw no internal structure (treating the atom as a solid ball — Dalton-level model); students who draw a nucleus with electrons somewhere around it (partial Rutherford knowledge); students who draw electrons in rings (suggesting prior exposure to Bohr model). This

📖 **PDF:** [topical-test-form-1-chemistry-introduction-to-chemistry-answers] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	teacher collects no books — this drawing stays with the student and will be revised at the end of Lesson 3 (Bohr model) and again at the end of Lesson 7 (orbital model). The lesson ends with the teacher previewing the next lesson: 'Next lesson, we go back to the beginning — to the scientists who first figured out what atoms actually look like. We start with a question: if you cannot see an atom, how do you figure out what is inside it?'	exactly how much your thinking has grown.' WAIT TIME: Allow the full 4 minutes for silent individual drawing — circulate and observe without commenting. After 4 minutes: 'Keep this drawing safe. Do not erase it. It is your scientific starting point.' Teacher closes the lesson: 'Today you did something important. You looked at a real phenomenon — one you may have seen at home without noticing it — and you asked scientific questions about it. You could not answer them yet. That discomfort you feel right now — that is what real scientific inquiry feels like. It is the best possible place to start. Next lesson, we begin building the tools to answer your questions.' Teacher gestures to the DQB: 'Those questions are waiting for us.'	misconceptions (e.g., students who draw fire going into a solid ball atom with no internal structure) that the teacher can specifically address in Lesson 2 when introducing Rutherford and Bohr models.	three-way classification tells the teacher how much scaffolding is needed at the start of Lesson 2. No marks are given — this is entirely diagnostic and motivational.

D. TEACHER REFLECTION
After the lesson, the teacher should reflect on four questions: (1) Did every student produce a prediction, an observation record, a two-column compare table, a wonder question on the DQB, and an initial model? If not, which students need individual check-ins at the start of Lesson 2? (2) What were the most generative student questions on the DQB — which ones reveal the deepest thinking and which reveal the most significant misconceptions? Plan to explicitly reference those specific questions by name in Lessons 3 and 7. (3) Did the cultural framing of the Kenyan cooking hearth story land authentically? Were students engaged or did it feel forced? Adjust the narrative in future iterations to include more student voices — for example, asking at the start whether any student has actually seen a colour change in a fire at home. (4) Was the wait time honoured? The 2-minute silences and 10-second observation pauses are not wasted time — they are the moments when the puzzle deepens in students minds. If the wait felt uncomfortable and was shortened, practise holding it in the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that sodium chloride produces a vivid yellow-orange flame, potassium permanganate produces a soft violet flame, and copper sulphate produces a blue-green flame, and that these colours are repeatable and specific to each substance.
What did I learn?	Students learned that different substances produce characteristic, repeatable colours in a flame, that this phenomenon is observable both in a school laboratory and in everyday Kenyan life such as at a cooking hearth, and that the colour is linked to the identity of the substance rather than to the colour of the solution or the temperature of the flame.
How does this explain the phenomenon?	Students cannot yet explain why different substances produce different colours — this productive gap is the engine of the entire unit. Initial ideas on the DQB reflect intuitive thinking rather than scientific models, and the initial atom drawings reveal baseline

	mental models that will be systematically developed across Lessons 2 through 9.

LESSON 2 (80 minutes): What Is an Atom? Reviewing Atomic Structure — Dalton & Rutherford Models

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To review the sub-atomic particles and the Rutherford nuclear model of the atom, and to begin building the explanation for why different atoms produce different flame colours.
Knowledge	Students will know the definitions of atomic number and mass number; can calculate numbers of protons, neutrons, and electrons; and can describe the key features and limitations of the Rutherford nuclear model.
Skills	Students will practise extracting and comparing data from element cards, interacting with a simulation to collect evidence, collaboratively constructing a physical atomic model, and recording model limitations.
Attitudes	Students develop curiosity about the internal structure of matter and appreciate that scientific models are revised when evidence reveals limitations.
Key Inquiry Question	How are electrons arranged in an atom?
Purpose in Storyline	This lesson re-activates prior knowledge of sub-atomic particles and introduces the Rutherford nuclear model, establishing that atoms have a tiny, dense nucleus with electrons somewhere outside it. This sets up the central unresolved question — where exactly are those electrons, and how does their position explain flame colours — which drives Lessons 3 through 9.
Safety Notes	Bunsen burners are not used in this lesson. No significant hazards are associated with printed element cards, the ARES offline simulation, bottle tops, or maize seeds. Ensure maize seeds are not placed in mouths or eyes, particularly where younger siblings may be present in open-lab settings.

B. LESSON OVERVIEW

In this 80-minute lesson, students revisit prior knowledge of atomic structure through a structured peer discussion using printed Element Cards for the first 20 elements (hydrogen to calcium). They begin by making individual predictions about sub-atomic particle counts before any peer discussion takes place, activating and surfacing existing mental models. They then gather first-hand evidence by comparing element cards in groups and interacting with the Rutherford Gold Foil Experiment simulation on the ARES platform — observing the three possible outcomes for alpha particles and recording the surprising deflection data that forced scientists to revise Dalton's solid-sphere model. In the Explain Phase, students make sense of their simulation observations: they construct written explanations for the deflection results and formalise the key vocabulary and relationships of the Rutherford nuclear model. The lesson is grounded in Kenya-relevant contexts throughout: the nucleus is introduced using the metaphor of the main house (the homestead centre) around which everything else is arranged, and the element cards feature elements prominent in Kenyan agriculture and daily life — potassium, sodium, calcium, and iron. Students then return to the class Driving Question Board (DQB) established in Lesson 1 to post new evidence cards and begin answering which of their original wonder questions Rutherford's model can and cannot resolve. Finally, students construct a physical Rutherford model using bottle tops as the nucleus and maize seeds as electrons, label its parts, record in their notebooks the model's key limitation — it cannot yet explain why different atoms produce different flame colours — and connect this gap explicitly to the driving question, preparing the ground for Lesson 3 on Bohr's model and energy levels.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Rutherford Scattering Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Rutherford's Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student receives a printed Element Card for one of the first 20 elements — for example sodium, potassium, calcium, or iron, all elements prominent in Kenyan daily life and agriculture. Working individually and in silence for 3 minutes, students fill in a personal 'What I Predict' table in their notebooks with columns for: atomic number, mass number, number of protons, number of neutrons, and number of electrons. They also sketch a rough diagram of what they think the atom looks like on the inside, labelling any parts they remember from prior learning. Students do not check textbooks or discuss with neighbours during this phase — the aim is to surface their current mental model honestly. At the end of the 3 minutes, students write one sentence completing the prompt: 'I think the different flame colours we saw in Lesson 1 might be related to atoms because…'	Distribute element cards before students open any books. Say: 'Do not look anything up yet — I want to know what is already in your head, not what is in your textbook. Work quietly for three minutes.' Circulate and observe without correcting; note two or three predictions — both accurate and inaccurate — that you will return to after the Observe Phase. After 3 minutes, say: 'Keep your predictions visible — we will come back to check them very shortly.' Do not confirm or deny any prediction at this stage. Allow a further 30 seconds of wait time after giving the writing prompt before moving on.	Individual prediction task activates prior knowledge and creates a personal benchmark that students will revise after gathering evidence, making conceptual change visible and concrete.	Teacher circulates and mentally notes which students confuse atomic number with mass number, or who cannot distinguish proton count from neutron count — these misconceptions are targeted during the Observe Phase group discussion.
Observe Phase 📹 **VIDEO:** Video: Rutherford Scattering Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Rutherford's Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Element Card Comparison (15 minutes): Students form groups of four. Each group member holds a different Element Card. Groups complete a shared 'Element Card Comparison Table' with columns: Element Name, Symbol, Atomic Number, Mass Number, No. of Protons, No. of Neutrons, No. of Electrons, and a final column labelled 'Anything unusual?' Each student fills in the row for their own element, then the group discusses: What patterns do you notice? Which numbers are always equal? Can you write a rule for calculating neutron number? Groups record any observations	For Part A, assign group roles (recorder, reporter, timekeeper, card-holder) before students sit together. After 8 minutes of group work, pause the class and ask one group to share a pattern they noticed — do not evaluate it yet, simply say 'Interesting — keep that thought.' For Part B, demonstrate the simulation controls on the projector for 60 seconds before students begin. Circulate and ask pairs: 'What surprised you about those results?' and 'What would you have expected if the atom were a solid ball?' Allow 2 minutes of wait time after pairs finish recording before calling the class	First-hand data collection from the simulation mirrors the process Geiger and Marsden used in 1909, helping students understand why the nuclear model was a revision of Dalton's model rather than an arbitrary replacement.	Check pairs' Results Tables for the Rutherford simulation: students who record that most particles pass straight through but cannot explain what that implies about atomic structure are flagged for targeted questioning in the Explain Phase.

	that surprised them in the 'Anything unusual?' column. PART B — Rutherford Simulation (20 minutes): Students open the Rutherford Gold Foil Experiment simulation on the ARES platform (available offline). Working in pairs, they fire alpha particles at the gold foil and record the three possible outcomes in a Results Table: (1) particle passes straight through, (2) particle deflects slightly, (3) particle bounces almost straight back. Pairs record the approximate frequency of each outcome and answer two observation questions in their notebooks: What did most of the particles do? What did the very small number of bounced-back particles tell the scientists?	back together. Do not explain the deflection yet — reserve that for the Explain Phase.		
Explain Phase 📹 **VIDEO:** Video: Rutherford Scattering Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Rutherford's Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students work in their groups of four to construct a written explanation for the simulation results they observed in the previous phase. Each group is given a structured 'Explanation Frame' with three prompts: (1) Most alpha particles passed straight through the gold foil because… (2) A small number were deflected at large angles because… (3) This means the atom must be mostly… with a tiny, dense… at its centre. Groups discuss and complete the frame, then share with the class. The teacher formalises key vocabulary on the board: nucleus, proton, neutron, electron, atomic number, mass number, and nuclear model. The teacher then introduces the Kenyan homestead metaphor: just as a traditional homestead has a main house (the nucleus) that is small but central, with the surrounding space mostly empty	Begin by asking: 'Look at your Results Table from the simulation. If the atom were a solid ball of positive charge — Dalton's model — what would you expect to happen to every single alpha particle?' Give 1 minute of wait time for pairs to discuss. Then ask: 'So what do the few particles that bounced back tell us must exist inside the atom?' After groups share their Explanation Frames, consolidate on the board using the three prompts as a structure. Introduce the homestead metaphor verbally: 'Think of the nucleus as the main house in a homestead — small, solid, central. The rest of the compound is mostly open ground. That is the Rutherford atom.' Write the key vocabulary on the board as each term is used. End the phase by asking: 'What one thing does Rutherford's model still not explain about the flame colours from	The structured Explanation Frame scaffolds evidence-to-explanation reasoning, ensuring students connect their observed data explicitly to the theoretical claim rather than simply accepting the nuclear model as received knowledge.	Collect Explanation Frames at the end of this phase. Check that students correctly link the large-angle deflections to the existence of a dense, positively charged nucleus rather than explaining them as random scatter.

	and the other structures (electrons) arranged in the open ground around it — so too is the atom mostly empty space. Students write a definition of the Rutherford nuclear model in their own words and note one thing the model explains well and one thing it cannot yet explain: why atoms of different elements produce different flame colours.	Lesson 1?' Wait 90 seconds for written responses before moving on.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Rutherford Scattering] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Rutherford's Atomic Model (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board created in Lesson 1. Each group receives two sticky cards in different colours: one colour for 'Evidence we now have' and one for 'New questions this raises'. Groups discuss and write: on the Evidence card, one clear statement about what Rutherford's model tells us about atomic structure; on the Question card, one question that Rutherford's model still cannot answer — particularly in relation to the anchoring phenomenon (flame colours). Cards are posted on the DQB under the relevant columns. The class then does a 3-minute gallery read of all newly posted cards. Students individually write in their notebooks: 'Rutherford showed us that… but the driving question still requires us to find out…' This consolidates the lesson's contribution to the overall unit storyline and makes explicit that today's model is a step forward but not yet a full answer. Students also return to their Phase 1 predictions and draw a line under their original entries, annotating what has changed in their thinking.	Direct students to the DQB physically — if it is on a wall, ask students to stand and walk to it rather than reading from their seats, so the ritual of posting feels deliberate. After cards are posted, facilitate the gallery read by saying: 'Read every new card — then return to your seat and write your sentence.' After the gallery read, ask one or two students to share their sentence aloud. Highlight any card that identifies the electron arrangement problem: 'This group has found exactly the gap we will investigate in Lesson 3.' Do not answer the electron-arrangement question now — record it on the DQB and say: 'Hold that question — it is the heart of this unit.'	Returning to the DQB makes the lesson's contribution to the unit storyline visible and tangible, reinforcing that each lesson is one step in a cumulative explanation rather than a stand-alone topic.	Read the Question cards posted on the DQB. Students who post questions about electron position or why different elements behave differently are demonstrating readiness for Lesson 3. Students whose questions remain at the level of particle counting may need additional support connecting structure to behaviour.
Model Building	Each group of four receives: 10 bottle tops (representing the	Distribute materials efficiently — pre-count bottle tops and maize	Constructing a physical model with everyday Kenyan materials (bottle	Assess the manila-paper models for three criteria: nucleus is small

Phase				
📹 **VIDEO:** [Video: Rutherford Scattering] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Rutherford's Atomic Model (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	nucleus — dense, small, concentrated mass) and 20 maize seeds (representing electrons — lightweight, surrounding the nucleus), plus a sheet of manila paper. Groups construct a physical Rutherford model: they cluster the bottle tops tightly at the centre of the manila paper to represent the nucleus, then arrange the maize seeds loosely in the large surrounding space to represent electrons. Groups label their model with arrows and annotations: nucleus (protons and neutrons), electrons, and the mostly empty space between them. They also write directly on the manila paper: 'What this model explains:' and 'What this model cannot yet explain:' — with the second prompt specifically pointing back to the driving question. Each group nominates a spokesperson to present their model to an adjacent group in a paired gallery share (2 minutes per group). Students then make a sketch of the completed model in their notebooks, ready to compare with the more developed models they will build in Lessons 3 and beyond.	seeds into small bags before the lesson. Say: 'Your model has to show three things: the nucleus is small and dense, the surrounding space is mostly empty, and the electrons are somewhere in that space.' Circulate and prompt groups that place electrons neatly in a ring: 'Rutherford did not actually know where the electrons were — does your model show that uncertainty?' After the paired gallery share, bring the class together and ask: 'What does every group have written under Cannot yet explain? Why is that the same for everyone?' End the lesson by saying: 'That gap is exactly what we will investigate in Lesson 3 — Bohr will take us one step closer to answering the driving question.' Allow 1 minute of silent reflection before students close their notebooks.	tops and maize seeds) makes the abstract nuclear model concrete and memorable, while the explicit limitations annotation ensures students do not mistake the model for a complete explanation.	and centralised, surrounding space is represented as mostly empty, and the 'cannot yet explain' annotation correctly identifies the unresolved electron-arrangement or flame-colour question.

D. TEACHER REFLECTION

After the lesson, consider: Did students move from surface-level recall (counting protons and neutrons) to genuinely puzzling over the electron-arrangement gap? Were the DQB question cards rich enough to drive curiosity into Lesson 3, or did most groups simply restate what Rutherford showed rather than identifying what remains unexplained? If the simulation was unavailable due to device issues, note whether the alternative (teacher demonstration with a projected image of the scattering pattern) produced equivalent conceptual engagement. Also reflect on whether the homestead metaphor for the nucleus resonated across different class contexts — urban students in Kisumu may need an adapted spatial metaphor.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed, via the Rutherford Gold Foil simulation on the ARES platform, that most alpha particles passed straight through

	the gold foil with little or no deflection, a small number deflected at moderate angles, and a very small number bounced back at large angles. They also observed, through the Element Card Comparison activity, that an element's atomic number equals its proton count, and that neutron number is calculated by subtracting atomic number from mass number.
What did I learn?	Students learned the definitions of atomic number, mass number, proton, neutron, and electron, and how to calculate sub-atomic particle counts. They learned the key features of the Rutherford nuclear model: an atom is mostly empty space, with a tiny, dense, positively charged nucleus at the centre and electrons somewhere in the surrounding space. They were introduced to the homestead-centre metaphor for the nucleus.
How does this explain the phenomenon?	The Rutherford model explains why atoms of different elements have different proton counts — a fact that is the same as having different atomic numbers — but it cannot yet explain why different elements produce different flame colours. The model establishes that electrons exist outside the nucleus in mostly empty space, creating the central unresolved question for the unit: where exactly are those electrons, and how does their specific arrangement account for the precise colours seen at the Kenyan cooking hearth? This gap motivates Lesson 3 on Bohr's model and energy levels.

LESSON 3 (80 minutes): Bohr's Model and Energy Levels	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce Bohr's planetary model of the atom and explain how electrons occupy specific energy levels (shells), connecting this structure to the observation that different substances produce different flame colours.
Knowledge	The learner illustrates the structure of the atom using Bohr's model and describes how electrons are arranged in discrete energy levels (shells) for the first 10 elements.
Skills	The learner draws accurate Bohr model diagrams for the first 10 elements and uses the model to predict electron behaviour when energy is absorbed from a flame.
Attitudes	The learner develops interest in the study of atomic structure by connecting Bohr's historical model to the culturally familiar phenomenon of flame colours observed at a Kenyan cooking hearth.
Key Inquiry Question	How are electrons arranged in an atom?
Purpose in Storyline	Lesson 3 introduces Bohr's planetary model as the first structured explanation for WHY different substances produce different flame colours. Having observed in Lessons 1–2 that (a) flame colours are substance-specific and repeatable, and (b) atoms have internal structure with a nucleus and electrons, learners now discover that electrons do not exist randomly — they occupy fixed energy levels. This is the critical mechanistic link: if electrons can only be at specific energy levels, then energy from the flame must cause discrete jumps, producing specific colours. Learners draw Bohr models for the first 10 elements and update the Driving Question Board with this new explanatory layer.
Safety Notes	No hazardous chemicals are used in this lesson. If a Bunsen burner is used for a brief teacher demonstration (optional recall activity), ensure the burner is on a stable heat-proof mat, hair is tied back, and loose clothing is secured. All students remain seated at a safe distance of at least 1 metre during any flame demonstration. Safety spectacles must be worn during any live flame activity.

B. LESSON OVERVIEW
Learners begin by recalling the flame colour observations from Lessons 1–2 and are challenged to explain WHY sodium always produces yellow light. The teacher introduces Niels Bohr's planetary model through a labelled diagram, establishing the concept of discrete electron energy levels (shells) with maximum electron capacities (2, 8, 8…). Learners draw Bohr models for hydrogen through neon (elements 1–10) in their exercise books. A structured Think-Pair-Share asks: "If electrons can only occupy certain energy levels, what must happen when an electron in a sodium atom absorbs heat energy from a cooking fire?" Learners articulate that electrons must jump to a higher level and, when they fall back, release energy as light — yellow light in sodium's case. The lesson closes with a DQB update in which learners add new explanatory sticky notes connecting Bohr's model to the anchoring phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are shown two images side by side on the board (or	Display the two images and read the prediction prompt aloud. Say:	Predict-Before-Instruction (Eliciting Prior Knowledge): By asking	Circulate during the 3-minute silent writing period. Look for: (a)

📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bohr's Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	printed handout): (1) a Kenyan cooking hearth with a yellow-orange flame where salt has been spilled, and (2) a simple diagram of an atom showing a nucleus with electrons scattered randomly around it (a 'cloud' — no shells). Learners individually write a 2-sentence response in their exercise books to the prompt: 'In Lessons 1 and 2 we saw that sodium always makes a yellow flame and potassium always makes a violet flame. Using what you know about atoms so far, why might the TYPE of atom inside a substance control the COLOUR of the flame it produces?' Learners are not expected to know the answer — predictions are explicitly welcomed as rough ideas. After 3 minutes of silent writing, learners share predictions with a shoulder partner for 2 minutes.	'You have seen this yellow flame at home — maybe when ugali salt hit the fire, or when someone cooked nyama choma. We are not guessing randomly — use what you already know about atoms from Lessons 1 and 2 to make your best scientific prediction. Write two sentences. Go.' Allow 3 minutes of SILENT individual writing. Do NOT accept shouted answers. After 3 minutes, say: 'Now turn to your shoulder partner. Share your prediction. You have 2 minutes.' After partner sharing, cold-call 3 learners using name cards or a roster — not volunteers. Ask each: 'What did you predict, and what in Lesson 1 or 2 gave you that idea?' After each response, allow 10 seconds of WAIT TIME before commenting. Record key prediction words on the board (e.g., 'electrons', 'energy', 'shells', 'nucleus') to revisit at the end of the lesson.	learners to commit to a written prediction BEFORE new content is introduced, the teacher surfaces existing mental models and creates cognitive dissonance. When Bohr's model is later introduced, learners actively compare the new information against their own prediction, deepening encoding. Recording prediction keywords on the board creates a visible anchor for reflection at the lesson's close.	learners who make no connection to atomic structure — these learners may need a brief Lesson 2 recap before proceeding; (b) learners who already use language like 'energy levels' or 'shells' — note these for strategic cold-calling during the Explain phase. Observable indicator of readiness: at least 70% of learners have written two complete sentences referencing atoms or electrons before the Observe phase begins.
Observe Phase 📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bohr's Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Guided Diagram Analysis (15 min): Each learner receives (or copies from the board) a printed/drawn Bohr model diagram of a sodium atom showing: nucleus (11 protons, 12 neutrons), Shell 1 (2 electrons), Shell 2 (8 electrons), Shell 3 (1 electron). A second diagram shows the same sodium atom with its outermost electron shown jumping UP to a higher shell (arrow labelled 'energy absorbed from flame') and then falling BACK DOWN (arrow labelled 'light energy released — yellow, 589 nm'). Learners annotate both diagrams using a provided vocabulary list: nucleus, electron, energy level / shell, ground state, excited state, photon / light	Distribute or project the two sodium Bohr diagrams. Say: 'Look carefully at Diagram 1. This is Niels Bohr's model of the sodium atom — the same sodium that is in the salt that makes a yellow flame at the cooking fire. What do you notice about where the electrons are sitting?' Allow 10 seconds WAIT TIME. Cold-call 2 learners. Then say: 'Now look at Diagram 2. Something has changed. An electron has moved. Where has it gone, and what do you think caused that?' Allow 10 seconds WAIT TIME. Cold-call 2 different learners. Introduce vocabulary explicitly: 'Bohr called these fixed paths energy levels or shells. Bohr's key idea — and this is the idea that will answer our driving	Annotated Diagram Analysis + Scaffolded Drawing Practice: Providing a pre-drawn but incomplete diagram (requiring annotation rather than passive reading) forces learners to actively process each structural feature. The sequential drawing task for elements 1–10 builds procedural fluency with the shell-filling rule and creates a personal reference set that learners will use in every subsequent lesson. Connecting the sodium jump diagram directly to the cooking hearth phenomenon ensures observation is always evidence for the driving question, not isolated content.	During the drawing practice, use a checklist to spot-check 8–10 learners' books. Key accuracy indicators: (1) Shell 1 never exceeds 2 electrons; (2) Shell 2 begins filling from lithium (Li = 2, 1); (3) Neon is drawn as 2, 8 with no partial shell. Common error to watch for: learners who place all electrons in one shell — redirect with: 'Count the spaces. Shell 1 is full at 2. Where does the next electron have to go?' Record names of learners with persistent errors for targeted support in Phase 3.

	released. Part B — Drawing Practice (15 min): Learners draw Bohr models independently for elements 1–10 (H, He, Li, Be, B, C, N, O, F, Ne) using the shell rules: Shell 1 max $2e^-$, Shell 2 max $8e^-$. They label each diagram with element name, symbol, atomic number, and number of electrons per shell.	question — is that electrons can ONLY exist at specific energy levels. They cannot be in between. Write that in your notes, underline it.' For the drawing practice, model the first two elements (H and He) on the board step by step, narrating: 'Hydrogen: atomic number 1, so 1 electron. Shell 1 has space for 2 electrons, so we place 1 electron in Shell 1. Done.' Then say: 'Now you complete elements 3 through 10. You have 15 minutes. I will be walking around. If you finish early, write one sentence next to each diagram: how many electrons are in the outermost shell?' Circulate continuously — check every learner's diagram for carbon ($6e^-$: 2, 4) and neon ($10e^-$: 2, 8) as checkpoint elements.		
Explain Phase 📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bohr's Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Think-Pair-Share (10 min): Learners respond to the structured prompt: 'A sodium atom is sitting in a cooking fire at a homestead in the Rift Valley. The flame gives the outermost electron in Shell 3 a burst of heat energy. Using Bohr's model, describe in THREE steps what happens to that electron — and why we see a yellow colour in the flame.' Learners think silently (2 min), pair with a partner (3 min), then pairs share with the class (5 min). Part B — Connecting to the Driving Question (10 min): The teacher writes on the board: 'Different substances produce different colours because ___.' Learners complete this sentence in their notes using Bohr's model vocabulary. They then answer: 'Why would potassium produce a DIFFERENT colour (violet) than sodium (yellow)? Use Bohr's	For the Think-Pair-Share, say: 'This is the moment we connect Bohr's model to our cooking fire mystery. Think silently for 2 minutes — no talking. Write your three steps.' After 2 minutes: 'Now share with your partner. Try to agree on the three steps.' After 3 minutes: 'Let us hear from some pairs.' Cold-call 3 pairs (not volunteers). After each response, press with: 'You said the electron jumps up — what do you mean by UP? UP in what sense?' and 'You said it releases light — why would it release light and not heat or sound?' Allow 10 seconds WAIT TIME after each probing question. After all responses, say: 'Here are the three steps — copy these into your notes exactly': (1) The flame gives the electron ENERGY — it ABSORBS that energy. (2) The electron JUMPS to a higher energy level — we call this the	Think-Pair-Share with Probing Questions (Structured Argumentation): The three-step narrative prompt (absorb → jump → fall → light) scaffolds learners in constructing a causal chain — moving from observation (colour) to mechanism (electron energy transitions). The probing questions ('What do you mean by UP?') push learners from surface recall to conceptual precision. Connecting the explanation directly to the anchoring phenomenon sodium atom in a Rift Valley cooking fire ensures that Bohr's model is always understood as an explanatory tool, not abstract theory.	Collect and read 5–6 sentence completions ('Different substances produce different colours because ___') as learners write. Observable success indicator: the sentence must include at minimum (a) electrons, (b) specific/fixed energy levels, and (c) the idea that different atoms have different energy level spacings. Learners whose sentences reference only 'the type of atom' without mechanistic language should be prompted: 'Can you add what the ELECTRONS are doing inside the atom?' The electron arrangement task (Na to P) is a rapid accuracy check — correct answers: Na=2,8,1; Mg=2,8,2; Al=2,8,3; Si=2,8,4; P=2,8,5.

	model in your answer.' Part C — Worked Example — Electron Arrangements (5 min): Teacher models writing the electron arrangement of phosphorus (P, Z=15) as 2, 8, 5, connecting back to the Bohr diagram. Learners then write electron arrangements (not full diagrams) for elements 11–15 in their books.	EXCITED STATE. (3) The electron is unstable at the higher level — it FALLS BACK to the lower level, releasing the energy it absorbed as a PHOTON of LIGHT. The colour of the light depends on EXACTLY HOW MUCH energy is released — and that depends on the specific atom's energy level spacing. That is why sodium is always yellow and potassium is always violet.' For the sentence completion, say: 'Now finish this sentence in your own words using those three steps.' Cold-call 2 learners to read their sentences. For the electron arrangement extension, model phosphorus on the board: '15 electrons. Shell 1: 2. Shell 2: 8. That is 10. I have 5 left. Shell 3: 5. So phosphorus is 2, 8, 5.' Say: 'Write arrangements for sodium (11) to phosphorus (15) in your books. You have 5 minutes.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bohr's Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or cuts two small paper squares). On the FIRST sticky note (green = 'I now understand'), they write one thing Bohr's model has explained about the cooking fire phenomenon that they could not explain before today. On the SECOND sticky note (orange = 'I still wonder'), they write one new question this lesson has raised. Learners post both sticky notes on the class Driving Question Board under the heading 'Lesson 3 — Bohr's Model'. The class then does a brief 3-minute gallery walk to read five other learners' sticky notes, placing a small tick on any 'wonder' question they also have. The teacher identifies the two most-ticked 'wonder' questions and writes them prominently on the DQB as 'Questions to	Say: 'Our Driving Question is: Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside atoms? Bohr's model has given us a new piece of the answer today. On your GREEN note, write exactly what Bohr's model explains about our cooking fire. Be specific — don't just write 'electrons'. On your ORANGE note, write one question you now have that we have NOT yet answered.' Allow 4 minutes for writing. Then say: 'Post your notes on the board. Then do a gallery walk — you have 3 minutes. Put a tick on any ORANGE question you also want answered.' After the gallery walk, read the two most-ticked questions aloud and write them on the DQB. Say: 'These are	Driving Question Board — Cumulative Sense-Making Artifact: The DQB functions as a living, public record of the class's growing understanding. The green/orange distinction trains learners to metacognitively separate 'what I now know' from 'what I still need to explain' — a core scientific practice. The gallery walk and tick-voting makes the epistemic community visible: learners see that their questions are shared, normalising scientific uncertainty. The teacher's verbal narration of the DQB's growth across Lessons 1–3 explicitly models how scientific understanding builds incrementally from evidence.	Read all posted green sticky notes before the next lesson. Observable success indicator: green notes should reference Bohr's model mechanistically (e.g., 'electrons jump to higher shells and release coloured light when they fall back') rather than only descriptively (e.g., 'sodium makes yellow light'). Tally how many orange notes ask questions about SPECIFIC colours, wavelengths, or why only certain energy jumps are possible — these indicate learners are ready for the spectroscopy content in Lessons 4–5. Learners whose green notes remain purely descriptive should be individually followed up at the start of Lesson 4.

	Investigate in Lessons 4–9'.	now on our board as open questions. We will be building answers to these over the next lessons. Keep them in mind.' Point to the DQB: 'Look at how our board is growing — in Lesson 1 we asked WHY the fire changes colour. In Lesson 2 we learned atoms have a nucleus and electrons. Today, Bohr tells us the electrons are in specific shells — and when they absorb and release energy, they produce specific colours. Our model is getting more powerful.'		
Model Building Phase 📹 **VIDEO:** [De Broglie wavelength] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Bohr's Atomic Model (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners open to their cumulative atomic model diagram started in Lesson 1 (or begin one if absent). They add the following new features in a different colour (e.g., red pen): (1) Concentric circles around the nucleus representing Shell 1, Shell 2, Shell 3. (2) Electrons placed as dots on the correct shells for their chosen element. (3) A curved arrow showing an electron in the outermost shell jumping UP to a higher shell (labelled 'energy absorbed from flame — excited state') and a second curved arrow showing the electron falling BACK DOWN (labelled 'photon of light released — specific colour = fingerprint of this atom'). (4) A written caption at the bottom: 'Bohr's model shows electrons occupy fixed energy levels. When excited by flame heat, outer electrons jump to higher levels. When they fall back, they release light of a specific colour. This is why sodium always makes yellow light at a Kenyan cooking fire.' Learners who finish early add Bohr models for sodium AND potassium side by side, annotating: 'Sodium	Say: 'Take out your cumulative model from Lesson 1. We are going to make it much more powerful today. Use a different colour pen so you can see what you are adding. First, draw three concentric circles around your nucleus — these are Bohr's energy levels. Second, place your electrons on the correct shells using the rules we practised. Third — and this is the most important part — draw the jump. Show an electron absorbing energy from the flame and jumping up, then falling back and releasing a photon of coloured light.' Project a completed example model on the board or hold up a hand-drawn teacher model. Say: 'Your model should now be able to EXPLAIN — not just describe — why different substances make different flame colours. Write that caption at the bottom. If you finish, try adding sodium and potassium side by side and compare their jumps.' Allow 8–10 minutes. With 2 minutes remaining, say: 'Hold up your model. I want to see the three concentric circles, the electrons in the right shells, and the jump	Cumulative Model Revision (Progressive Model Building): The model functions as a persistent, learner-owned sensemaking artefact. By adding new features in a distinct colour each lesson, learners can visually track how their explanatory model has evolved from Lesson 1's basic 'nucleus + electrons' sketch toward a fully mechanistic quantum model by Lesson 9. The explicit caption-writing requirement forces translation of visual model features into precise scientific language, consolidating both the diagram and the explanation simultaneously. Comparing sodium and potassium models side-by-side (extension) seeds the understanding of spectral fingerprints needed in Lesson 5.	Rapid visual scan during the 'hold up your model' moment. Three non-negotiable observable indicators: (1) Three concentric shells are drawn around the nucleus. (2) Electrons are correctly distributed across shells (Shell 1 ≤ $2e^-$). (3) At least one jump arrow is present with labels for 'energy absorbed' and 'light released'. Learners missing any of these three features are flagged for a 5-minute targeted review at the start of Lesson 4. Collect 5 books at random at the end of class for written feedback — return at the start of Lesson 4.

	releases more/less energy than potassium per jump, so their photons have different colours.'	arrow. If those three things are on your diagram, you have successfully built today's model.' Do a rapid visual scan of the room. Identify 2–3 exemplary models and ask those learners to briefly hold their books up or describe their model to the class.		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions before planning Lesson 4:

1. PREDICTION QUALITY: Did learners' predictions in Phase 1 reveal any persistent misconceptions — for example, the idea that 'hotter flames make different colours' or that 'colour comes from the fuel/wood, not the substance added'? If so, plan a brief explicit misconception-address warm-up for Lesson 4.

2. BOHR MODEL ACCURACY: During the drawing practice (Phase 2), how many learners placed more than 2 electrons in Shell 1, or skipped shells? If more than 30% of the class made this error, revisit the shell-filling rules at the start of Lesson 4 using a whole-class error analysis before introducing Shell 3 and elements 11–18.

3. THINK-PAIR-SHARE DEPTH: Were learners able to articulate all three steps (absorb → excited state → fall back → photon released) in their own words, or did they reproduce the teacher's language without conceptual ownership? If most responses were rote, plan a Lesson 4 warm-up where learners must explain the three steps using a NEW Kenyan context — e.g., copper wire turning a flame green at a Kisumu school lab.

4. DQB ORANGE NOTES: What questions appeared most frequently? Questions about WHY specific atoms release specific COLOURS (not just any colour) indicate learners are ready for emission spectra in Lesson 5. Questions about WHAT an 'excited state' feels like physically suggest a need for a more concrete analogy (e.g., jumping up stairs: you need exactly the right energy to land on a step, not between steps).

5. MODEL BUILDING COMPLETENESS: Did the 'hold up your model' scan reveal any learners with incomplete or inaccurate jump arrows? These learners have the most critical gap — the jump mechanism IS the explanation for the phenomenon. Prioritise these learners for early support in Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Annotated Bohr model diagrams of sodium showing an electron in Shell 3 absorbing energy from a flame (Rift Valley cooking hearth), jumping to a higher energy level (excited state), then falling back to the ground state and releasing a yellow photon. Bohr model drawings for elements 1–10 (hydrogen to neon) showing electrons correctly distributed across shells (Shell 1: max $2e^-$; Shell 2: max $8e^-$). Electron arrangement notation for elements 11–15 written as comma-separated shell occupancies (e.g., Na = 2, 8, 1).
What did I learn?	Bohr's planetary model proposes that electrons occupy fixed, discrete energy levels (shells) around the nucleus — they cannot exist between shells. Shell 1 holds a maximum of 2 electrons; Shell 2 holds a maximum of 8 electrons. When a flame supplies energy to an atom (e.g., sodium in cooking salt), the outermost electron absorbs that energy and jumps to a higher energy level — this is the excited state. The excited state is unstable; the electron immediately falls back to its original (ground state) energy level, releasing the absorbed energy as a photon of light. The colour of that light is determined by the exact amount of energy released, which depends on the specific spacing of the atom's energy levels. Because each element has uniquely spaced energy levels, each element produces a unique, characteristic colour — sodium always yellow, potassium always violet. This is why the cooking fire at the Rift Valley homestead changes colour depending on WHICH substance is added to the flame.
How does this explain the phenomenon?	Bohr's model begins to explain the anchoring phenomenon: the reason different substances produce different flame colours is that different atoms have different electron energy level spacings. When thermal energy from the flame excites an electron to a higher

	level, the photon of light released as it falls back carries an amount of energy that is unique to that atom's energy level gap — producing a characteristic colour. Sodium's energy level spacing corresponds to yellow light (589 nm); potassium's spacing corresponds to violet light. This means flame colour is a direct observable fingerprint of atomic electron structure — connecting the culturally familiar cooking hearth observation to the internal quantum structure of atoms described by Bohr's model.

LESSON 4 (80 minutes): Isotopes and Relative Atomic Mass — Part 1: Meaning and Concept	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To develop learners' understanding of isotopes — what they are, how they differ, and crucially, why isotopes of the same element share identical electron arrangements and therefore produce the same flame colour.
Knowledge	Learners will know that isotopes are atoms of the same element with the same atomic number but different mass numbers (due to differing numbers of neutrons); that the relative atomic mass of an element is a weighted average of the masses of its naturally occurring isotopes; and that isotopic identity does not alter electron configuration or chemical behaviour.
Skills	Learners will be able to: (a) identify and distinguish isotopes using atomic number and mass number notation; (b) explain why isotopes of the same element produce the same flame colour; (c) construct labelled nuclear composition diagrams for isotopes of hydrogen (protium, deuterium, tritium) and chlorine (Cl-35, Cl-37); (d) use carbon-14 dating as a real-world application context for isotopes.
Attitudes	Learners will develop interest in atomic structure by appreciating how isotopic science connects to Kenyan heritage (carbon-14 dating of archaeological sites); advocate harmonious group discussion (social justice value); and gain confidence in presenting findings on isotopes (self-esteem PCI).
Key Inquiry Question	How are electrons arranged in an atom?
Purpose in Storyline	Lessons 1–3 established atomic number, mass number, and the Bohr model of electron energy levels. Learners can already say that sodium's electrons — excited by the flame — release yellow light when they fall back to ground state. Lesson 4 introduces a vital nuance: nature provides more than one 'version' of most atoms (isotopes). By discovering that isotopes share the same atomic number — and therefore the same electron arrangement — learners cement the rule that ELECTRON ARRANGEMENT, not nuclear mass, determines flame colour and chemical behaviour. This directly advances the driving question and clears a potential misconception before Lesson 5 (relative atomic mass calculations).
Safety Notes	No hazardous chemicals are used in this lesson. Learners handle paper-based and digital materials only. Remind learners to handle tablets/devices responsibly (Integrity value). Ensure group furniture is arranged safely to allow movement during the gallery walk.

B. LESSON OVERVIEW

Learners begin by revisiting the anchoring phenomenon — why sodium always gives yellow and potassium always gives violet — and are challenged with a new puzzle: scientists have discovered that chlorine atoms from the ocean near Mombasa do not all have the same mass, yet seawater burns with only one characteristic chlorine-family colour. Through a structured brainstorm, learners co-construct the concept of isotopes, then examine nuclear composition diagrams for hydrogen and chlorine isotopes. A carbon-14 dating context (Kenyan archaeological sites at Olorgesailie and Kariandusi) grounds the learning in local heritage. Learners update the Driving Question Board and close by revising their atomic model to show that isotopes share electron arrangement.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown a provocative	Display the two envelope visual	Individual Written Prediction with	Observable indicator: Each

📹 **VIDEO:** [Second Law of Thermodynamics and entropy] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Dalton's Atomic Theory (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	display prompt projected on the board: two sealed envelopes labelled 'Chlorine Atom A — mass 35' and 'Chlorine Atom B — mass 37', alongside a photograph of a coastal dhow fire in Mombasa harbour burning driftwood soaked in seawater. The teacher poses: 'Both atoms are chlorine. One is heavier. When both are placed in a flame — do they glow different colours, or the same colour? Write your prediction and reason in your exercise book. You have 2 minutes — work alone.' Learners write individual predictions (same colour / different colour) with a justification rooted in what they already know about atomic number, mass number, and electron energy levels from Lessons 1–3.	and the Mombasa dhow photograph on the board. Read the prompt aloud. Say exactly: 'Remember what we established in Lesson 3 — the colour of the flame is determined by how electrons in an atom absorb and release energy. Now use that idea to make your prediction. Do NOT discuss with your neighbour yet.' Set a visible 2-minute timer. Circulate silently — scan for the range of predictions (expect roughly 50–60% predicting 'same colour' and 40–50% predicting 'different colour'). After time is up, cold-call 4 learners: 2 who predict 'same', 2 who predict 'different'. For each, ask: 'What piece of evidence from our DQB made you think that?' WAIT 12 seconds after each question before accepting responses. Do not confirm or correct — record both predictions on the board under columns headed SAME and DIFFERENT. Say: 'We will return to this at the end of today's lesson to see who was right — and more importantly, why.'	Evidence Anchoring — learners are required to link their prediction to prior lesson evidence (electron energy levels), activating schema from Lessons 1–3 and exposing existing mental models about what determines flame colour. Recording predictions publicly creates cognitive dissonance that motivates the lesson's investigation.	learner's exercise book contains a written prediction AND a written justification that references at least one atomic concept (atomic number, electron arrangement, or energy levels). Teacher scans books during circulation — flag learners who write only 'same' or 'different' without justification for targeted questioning during Phase 3. Tally of board predictions reveals class-wide prior conception for teacher to address explicitly.
Observe Phase 📹 **VIDEO:** [Second Law of Thermodynamics and entropy] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Dalton's Atomic Theory (Flexbook)] Source: CK-12 🔍 Search ARES	Activity A — Carbon-14 Context Card (10 min): Each group of 4 receives a laminated Context Card describing the Olorgesailie Prehistoric Site in the Rift Valley. The card explains: 'Kenyan archaeologists use carbon-14 dating to determine the age of ancient tools found at Olorgesailie. Carbon-14 is a special form of carbon. Normal carbon has a mass number of 12. Carbon-14 has a mass number of 14. Both are carbon — both have 6 protons. But carbon-14 has 2 extra neutrons.' Groups are then given a 'Nuclear Composition Table'	Distribute Context Cards and worksheets. Say: 'You have 10 minutes. Your job as a group is to fill in every cell of the nuclear composition table. Use the formula: number of neutrons = mass number − atomic number. The AZX notation cards on your desk are your reference.' Circulate to each group. After 3 minutes, if groups are stuck on deuterium, prompt: 'Deuterium has mass number 2 and atomic number 1. How many neutrons does that give?' WAIT 10 seconds. For the gallery walk, say: 'You are detectives. Each group has 2	Context Card + Nuclear Composition Table (Structured Data Collection) followed by a Gallery Walk with Colour-Coded Dot Annotation — the two-stage activity separates data gathering (filling the table) from pattern recognition (the gallery walk). The Kenyan archaeological context (Olorgesailie, Kariandusi) grounds an abstract concept in a culturally meaningful application, helping learners see isotopes as real and consequential rather than theoretical. The gallery walk leverages peer-generated data, making the pattern discovery	Observable indicator: Each group's completed Nuclear Composition Table is correct for at least 4 of the 5 isotopes listed (C-12, C-14, ^{1}H, ^{2}H, ^{35}Cl). Teacher uses the gallery walk dot distribution as a quick diagnostic — if red dots appear on the 'protons' column (meaning learners think proton number changes in isotopes), this is a critical misconception to address immediately at the start of Phase 3. Collect one table per group at the end of Phase 2.

for similar readings	worksheet and must fill in: number of protons, neutrons, and electrons for C-12 and C-14. They also complete the same table for hydrogen isotopes (protium: ^{1}H, deuterium: ^{2}H, tritium: ^{3}H) and chlorine isotopes (^{35}Cl, ^{37}Cl) using the AZX notation cards provided. Activity B — Spot-the-Pattern Gallery Walk (10 min): Completed tables from each group are posted on the classroom walls. Groups rotate (2 minutes per station) and use sticky dots to mark any pattern they notice. The key pattern learners are expected to identify: 'The number of protons (atomic number) is always the same for isotopes of the same element — but the number of neutrons changes.'	minutes at each table. Place a green dot next to anything that is the SAME across isotopes of an element. Place a red dot next to anything that CHANGES.' After the walk, bring the class back and cold-call 3 groups to share one pattern each. WAIT 12 seconds between each response. Write the emerging class consensus on the board: 'Isotopes = same atomic number (Z), different mass number (A), different neutron count.' Do NOT yet state the electron arrangement implication — save that for Phase 3.	collaborative and visible.	
Explain Phase 📹 **VIDEO:** Second Law of Thermodynamics and entropy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Dalton's Atomic Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Activity A — Connecting to Electrons (8 min): Learners return to their groups. The teacher projects a side-by-side comparison of ^{35}Cl and ^{37}Cl: nuclear composition diagrams showing protons, neutrons, and — crucially — the electron arrangement shells (2, 8, 7 for both). Learners are asked: 'Look at the electron arrangement of Cl-35 and Cl-37. What do you notice? What does this mean for the flame colour test?' Groups discuss for 4 minutes, then each group completes a 'Connect the Dots' sentence frame: 'Cl-35 and Cl-37 are isotopes because ___. Their electron arrangement is ___ because ___. This means that in a flame test, they would produce ___ colour because ___.' Activity B — Isotope Sort Card Game (7 min): Each group receives a set of 12 atom cards (printed on cardstock). Each card shows: element symbol,	For Activity A, project the dual Cl-35/Cl-37 diagram. Say: 'I want you to look very carefully at the BOTTOM of each diagram — the electron shells. Count the electrons in each shell for both isotopes.' WAIT 15 seconds. Cold-call 2 learners: 'What did you count?' Then ask the whole class: 'If the electron arrangement is identical, and flame colour depends on electron energy levels — what must be true about the flame colour of Cl-35 versus Cl-37?' WAIT 12 seconds. Cold-call 3 learners. Guide toward the class consensus: 'Same electron arrangement → same energy levels → same flame colour, regardless of neutron count.' For Activity B, circulate during the card sort. Challenge fast-finishing groups: 'You have ^{23}Na and ^{24}Na in your pile. Sodium-23 is stable; sodium-24 is radioactive and used in medical tracers here in Kenya.	Sentence Frame Completion (Claim-Evidence-Reasoning scaffold) + Isotope Sort Card Game — the sentence frame forces learners to make an explicit logical chain from isotope definition → electron arrangement → flame colour, directly answering the driving question. The card sort is a kinaesthetic classification task that consolidates the definition through repeated application across multiple elements. Both strategies require learners to use the anchoring phenomenon as the explanatory endpoint, preventing surface-level memorisation.	Observable indicator: Each group's completed sentence frame correctly states (a) why Cl-35 and Cl-37 are isotopes (same Z, different A), (b) that electron arrangement is identical (2,8,7), and (c) that flame colour would be the same. Teacher reviews 2–3 sentence frames per group during circulation. For the card sort, the critical error to watch for is learners sorting by mass number instead of atomic number — intervene immediately with a pointed question: 'Which number tells you the element's identity — Z or A?'

	atomic number, mass number. Groups must sort them into isotope families. Cards include: ^{1}H, ^{2}H, ^{3}H / ^{12}C, ^{13}C, ^{14}C / ^{35}Cl, ^{37}Cl / ^{23}Na, ^{24}Na / ^{39}K, ^{40}K, ^{41}K. After sorting, groups must answer: 'Which of these elements appear in our anchoring phenomenon (the Kenyan cooking hearth)?' (Answer: Na, K, Cl — sodium gives yellow, potassium gives violet.) 'Do the different isotopes of sodium change the flame colour? Why or why not?'	Do they give the same flame colour? Prove it using atomic number.' After 7 minutes, cold-call one group to share their isotope families. Correct any misclassification immediately with: 'Remember — same element means same number of protons. Check atomic number Z first, always.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Second Law of Thermodynamics and entropy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Dalton's Atomic Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board, which has accumulated sticky notes, claims, and questions from Lessons 1–3. Each learner receives two sticky notes. On the YELLOW note, they write one new claim they can now make about isotopes and flame colour — something they could NOT have said at the start of today's lesson. On the BLUE note, they write one new question that today's lesson has raised (e.g., 'If isotopes of the same element have the same flame colour, how do scientists tell them apart in real experiments?'). Learners post their notes on the DQB under the column: 'Lesson 4 — Isotopes.' The class then does a 3-minute 'DQB Scan' — learners walk up, read three notes that are not their own, and give a thumbs-up to any claim they agree with. The teacher reads aloud 2 claims and 2 questions selected for accuracy and depth.	Hand out sticky notes. Say: 'Your yellow note must contain the word ISOTOPE and the phrase ELECTRON ARRANGEMENT. Your blue note must start with the word HOW, WHY, or WHAT. You have 3 minutes — write alone.' WAIT 3 minutes. Direct learners to post notes. During the scan, say: 'You are looking for a claim that is scientifically accurate and connects to the cooking hearth phenomenon. Give it a thumbs-up.' After the scan, read aloud 2 yellow claims and 2 blue questions. For each yellow claim, ask the class: 'Is this claim supported by the evidence we gathered today? Raise your hand if yes.' For each blue question, say: 'Keep this question in mind — some of you will find answers in Lesson 5, others in Lessons 6 and 7.' Point to the driving question displayed at the top of the DQB and say: 'We now know that flame colour is determined by electron arrangement — and isotopes do NOT change electron arrangement. We are getting closer to a complete answer.'	Driving Question Board — Claim and Question Addition with Peer Validation — the DQB is a cumulative public record of the class's evolving understanding. Requiring specific vocabulary (isotope, electron arrangement) on yellow notes ensures metacognitive precision. Blue question notes surface genuine curiosity and productive confusion, which the teacher can use to preview upcoming lessons. Peer thumb-voting creates low-stakes social accountability for scientific accuracy.	Observable indicator: At least 80% of yellow sticky notes correctly state that isotopes share the same electron arrangement and therefore produce the same flame colour in a flame test. Teacher photographs the DQB at the end of this phase for later review. Absence of the concept 'electron arrangement' on a yellow note flags a learner who has not yet made the critical connection — these learners are prioritised for check-in at the start of Lesson 5.
Model	Learners open to the atomic model	Say: 'Open your journals to your	Cumulative Model Revision with	Observable indicator: Each

Building Phase 📹 **VIDEO:** [Second Law of Thermodynamics and entropy] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Dalton's Atomic Theory (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	diagram they have been building cumulatively across Lessons 1–4 in their Science journals. Today's task: revise the model to show isotopes. Learners choose ONE element from the anchoring phenomenon (sodium, potassium, or chlorine) and draw TWO versions of its atom side by side — showing two of its naturally occurring isotopes. Labels required: atomic number (Z), mass number (A), number of protons, number of neutrons, number of electrons, and electron shell arrangement. Learners then draw a labelled arrow between the two isotopes and write: 'Same Z → same electron arrangement → same flame colour.' Below the diagrams, learners write a 2-sentence 'Model Revision Note' explaining what is NEW about their model compared to Lesson 3's model. Finally, learners revisit their Phase 1 prediction and write: 'My prediction was [correct/incorrect] because...' in a box at the bottom of the page.	cumulative atomic model. Today you are adding an isotope layer. Choose sodium, potassium, or chlorine — these are the stars of our cooking hearth story.' Display a model exemplar on the board showing ^{23}Na and ^{24}Na side by side with all required labels. Say: 'Your model must show two isotopes of your chosen element. Every label I have shown must appear on your diagram. You have 10 minutes.' Circulate and check: (1) Are both isotopes of the SAME element (same Z)? (2) Do both show the SAME electron shell arrangement? (3) Is the arrow and caption present? For learners who finish early, prompt: 'Can you add a third isotope? What would change — and what would stay the same?' With 2 minutes remaining, say: 'Now go back to your Phase 1 prediction on page one of today's notes. Write whether you were right or wrong — and use the word ELECTRON ARRANGEMENT in your explanation.' Cold-call 2 learners to share their prediction reflection aloud.	Prediction Reconciliation — the cumulative journal model is a central sensemaking artefact of the entire unit. By requiring learners to add isotope layers onto existing atomic models (rather than drawing new ones), this strategy makes conceptual growth visible and tangible. Returning to the Phase 1 prediction closes the lesson's cognitive loop, forcing learners to explicitly reconcile prior belief with new evidence — a hallmark of scientific thinking aligned with NGSS Science and Engineering Practice 6 (Constructing Explanations).	learner's journal model shows two correctly labelled isotopes of the same element (same Z, different A, same electron arrangement), with the arrow-caption linking same electron arrangement to same flame colour. The prediction reconciliation box contains the phrase 'electron arrangement' and a causally accurate explanation. Teacher collects 6 journals (2 high, 2 mid, 2 developing) for detailed review before Lesson 5. Any journal missing the electron arrangement caption is flagged for a targeted opening warm-up in Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Were learners able to articulate — in their own words — WHY isotopes of the same element produce the same flame colour? Look for this specifically in the Model Building phase journals. (2) During the gallery walk in Phase 2, were there groups who placed red dots on the 'protons' column, suggesting the misconception that isotopes have different atomic numbers? If so, plan a brief warm-up for Lesson 5 in which learners physically sort 5 atom cards by atomic number before anything else. (3) Were the carbon-14 / Olorgesailie context cards engaging enough to ground abstract isotope language in Kenyan reality? Gather verbal feedback from 3–4 learners at the door: 'Tell me one thing you found interesting about carbon-14 dating in Kenya.' (4) How evenly did groups distribute speaking turns during the card sort and gallery walk? Note any groups where one learner dominated — adjust groupings for Lesson 5 if needed. (5) Review the blue DQB sticky notes: are learners asking questions that naturally lead to relative atomic mass calculations (Lesson 5) and electron configuration (Lessons 6–8)? If questions cluster around radioactivity rather than electron arrangement, reorient the Lesson 5 opening to explicitly bridge back to the driving question.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Groups completed Nuclear Composition Tables for C-12, C-14, ^{1}H, ^{2}H, ^{3}H, ^{35}Cl, and ^{37}Cl using AZX notation. During the Gallery Walk, learners used green and red sticky dots to identify what stays the same (proton number / atomic number) and what changes (neutron count / mass number) across isotopes of the same element. Learners also sorted 12 atom cards into isotope families, identifying sodium, potassium, and chlorine as elements from the anchoring Kenyan cooking hearth phenomenon.
What did I learn?	Isotopes are atoms of the same element that have the same atomic number (same number of protons and electrons) but different mass numbers (different numbers of neutrons). Because atomic number is unchanged, isotopes share identical electron arrangements. Since flame colour is produced by electron energy transitions — not by nuclear mass — isotopes of the same element always produce the same flame colour. Carbon-14, used in dating Kenyan archaeological sites like Olorgesailie, is an isotope of carbon-12: same 6 protons, but 2 extra neutrons.
How does this explain the phenomenon?	Isotopes of the same element produce the same colour in a flame test because flame colour is determined by the electron arrangement of an atom, which depends on atomic number (number of protons). Since isotopes share the same atomic number, their electrons occupy the same energy levels. When heated, those electrons jump to the same higher energy levels and release the same wavelengths of light when falling back — producing an identical colour. Mass number (and therefore neutron count) does not affect electron arrangement, so it does not affect flame colour.

LESSON 5 (80 minutes): Isotopes and Relative Atomic Mass — Part 2: Calculation and Weighted Averages

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to calculate the relative atomic mass of elements from isotopic abundance data using the weighted average formula, and to consolidate understanding that atomic identity — and therefore flame colour — is determined by atomic number, not atomic mass.
Knowledge	Learners will know that: (a) relative atomic mass is a weighted average of the masses of an element's isotopes, taking into account the percentage abundance of each isotope; (b) the weighted average formula is RAM = Σ(isotope mass × fractional abundance); (c) different isotopes of the same element have the same atomic number and the same electron arrangement, and therefore produce the same flame colour.
Skills	Learners will be able to: (a) correctly apply the weighted average formula to calculate relative atomic mass from given isotopic abundance data; (b) present and justify their calculations clearly to peers during a gallery walk; (c) evaluate the reasonableness of a calculated RAM by checking it falls between the lightest and heaviest isotopes.
Attitudes	Learners will: (a) demonstrate self-esteem and confidence when presenting computed values to peers (Life Skills PCI); (b) advocate harmonious group collaboration consistent with the value of Social Justice; (c) show integrity by showing all working steps and not copying results.
Key Inquiry Question	How are electrons arranged in an atom?
Purpose in Storyline	In Lesson 4, learners discovered that isotopes are atoms of the same element with different numbers of neutrons, and that all isotopes of an element share the same atomic number and electron arrangement. Lesson 5 advances this by giving learners the mathematical tool — the weighted average — to calculate relative atomic mass from real isotopic abundance data for chlorine and carbon. By computing a single average mass from a mixture of isotopes, learners deepen the key storyline insight: because all isotopes of sodium share atomic number 11 and identical electron configurations, they ALL produce the same brilliant yellow-orange flame regardless of their mass. Atomic mass does not determine flame colour — atomic number does. This lesson updates the Driving Question Board accordingly and positions learners to move next into electron arrangement in energy levels.
Safety Notes	No hazardous materials are used in this lesson. Ensure gallery-walk posters are secured safely to walls or boards. If scissors are used to cut data cards, supervise carefully. Remind learners to handle printed resource sheets with care and return all materials after the lesson.

B. LESSON OVERVIEW

Learners open with a quick predict-phase estimation of chlorine's atomic mass from memory, then receive printed isotopic abundance data cards for chlorine (Cl-35: 75%, Cl-37: 25%) and carbon (C-12: 98.9%, C-13: 1.1%). Through a structured worked-example and guided practice, they derive and apply the weighted average formula to calculate relative atomic mass. Groups then prepare mini-posters of their calculations and present them in a gallery walk, building the Life Skills PCI of self-esteem. The lesson closes with a whole-class sensemaking discussion linking the calculated RAM back to the anchoring phenomenon: because isotopes of the same element share the same electron arrangement, they produce identical flame colours — confirming that flame colour is a fingerprint of atomic number, not atomic mass. The Driving Question Board is updated to reflect that RAM is a weighted average and that element identity is governed by proton number.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Isotopes and Atomic Mass Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Dalton's Atomic Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually write a prediction on a sticky note: 'The periodic table lists chlorine's atomic mass as approximately 35.5. But we know chlorine only exists as Cl-35 and Cl-37 — both whole numbers. How can the atomic mass be 35.5? Write your best explanation in one sentence before we investigate.' Learners hold up their sticky notes simultaneously on the teacher's count of three. Selected predictions are read aloud by the teacher and pinned to the Driving Question Board under the column 'What we THINK we know.'	Display the prompt on the board: 'Chlorine's RAM on the periodic table is 35.5. Cl-35 and Cl-37 are both whole numbers. How is 35.5 possible?' Say exactly: 'Do not look at your neighbour's note yet. Write your own best thinking — there is no wrong answer at this stage. You have 90 seconds.' After 90 seconds, say: 'Hold up your sticky note on three — one, two, three!' Cold-call 4 learners to read their predictions aloud. Ask: 'Who predicted it has something to do with a mixture or average?' (Show of hands.) Say: 'Keep that idea — we are going to test it with real data today.' Pin all sticky notes to the DQB.	Prediction Journaling with Sticky Notes — by committing a written prediction before instruction, learners activate prior knowledge about isotopes from Lesson 4 and create a personal cognitive anchor. The public display of predictions without judgment builds psychological safety and surfaces the range of ideas in the room.	Observable indicator: All learners produce a written prediction within 90 seconds. Teacher scans for the key idea of 'mixture' or 'average.' If fewer than 30% mention averaging or mixing, teacher notes this and plans to make the analogy to exam-score averages more explicit in Phase 2.
Observe Phase 📹 **VIDEO:** Video: Isotopes and Atomic Mass Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Dalton's Atomic Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a printed Data Card (one per pair) containing: (A) Chlorine isotope data — Cl-35: mass 35, abundance 75%; Cl-37: mass 37, abundance 25%. (B) Carbon isotope data — C-12: mass 12, abundance 98.9%; C-13: mass 13, abundance 1.1%. The teacher first projects a Kenya-context analogy: 'Imagine a maize farmer in Kericho harvests two varieties of maize — heavy cobs (mass 3 kg each) making up 75% of the harvest, and light cobs (mass 1 kg each) making up 25% of the harvest. The average cob mass is NOT simply (3+1)/2 = 2 kg. It is weighted: (3×0.75) + (1×0.25) = 2.5 kg.' Learners calculate the maize analogy first (2 minutes). The teacher then formally introduces the weighted average formula: RAM = ($mass_1$ × $abundance_1$/100) + ($mass_2$ × $abundance_2$/100). Learners work in pairs to calculate the RAM of	Distribute data cards face-down. Say: 'Turn over your card only when I say go.' After the maize analogy is projected, say: 'Calculate the weighted average cob mass. You have 2 minutes — go.' After 2 minutes, cold-call 2 pairs to give their answer. Confirm 2.5 kg with the class. Then say: 'Now we apply this exact same logic to atoms. The formula is RAM = ($mass_1$ × fractional $abundance_1$) + ($mass_2$ × fractional $abundance_2$). Let us write it together.' Write the formula on the board step-by-step. Say: 'For Cl-35 and Cl-37, substitute the numbers from your data card. Work with your partner. You have 4 minutes.' Walk the room — crouch beside at least 3 pairs and silently observe their written working without intervening unless a fundamental error is made. After 4 minutes, pause the class. Ask: 'What did you get?' Wait 10 seconds. Cold-call 3 pairs.	Concrete-Representational-Abstract (CRA) Bridging — the Kericho maize-harvest analogy provides a concrete, Kenya-relevant weighted average before the abstract atomic formula is introduced. This reduces cognitive load and makes the mathematical operation feel familiar rather than new. Pair work ensures every learner writes the calculation, not just observes it.	Observable indicator: Teacher scans exercise books during circulation — target indicator is that pairs correctly convert percentage to fractional abundance (dividing by 100) and show two multiplication steps before summing. Common error to watch for: learners simply averaging (35+37)/2 = 36 without weighting. If this error is seen in more than 3 pairs, teacher pauses the class and returns to the maize analogy.

	chlorine step-by-step, writing every line of working in their exercise books. The teacher circulates and observes written working.	Write their answers on the board. Facilitate a brief class vote on which answer is correct and why. Confirm RAM of chlorine ≈ 35.5. Ask: 'Does 35.5 lie between 35 and 37?' (Yes.) Say: 'Good — this is your reasonableness check. Your RAM must always fall between the lightest and heaviest isotope.' Repeat the calculation sequence for carbon as independent pair work (5 minutes).		
Explain Phase 📹 **VIDEO:** Video: Isotopes and Atomic Mass Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Dalton's Atomic Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four are formed (two pairs join). Each group receives a large sheet of paper and markers. Their task: prepare a Gallery Walk Poster that must include — (1) the weighted average formula written clearly, (2) the full step-by-step calculation for BOTH chlorine and carbon, (3) a 'Reasonableness Check' statement confirming the RAM lies between the isotope masses, (4) a one-sentence answer to: 'Why do Cl-35 and Cl-37 produce the same colour in a flame test, even though they have different masses?' Groups have 12 minutes to prepare their posters. Posters are then mounted on the classroom walls or desks in a gallery. Each group rotates to view two other groups' posters, leaving a sticky-note comment: one 'Star' (something done well) and one 'Wish' (a suggestion for improvement). Groups then return to their own poster and read the feedback they received.	Assign groups by proximity (no reshuffling needed). Say: 'Your poster must contain all four elements I am pointing to on the board — formula, full working for BOTH elements, a reasonableness check, and the flame-colour sentence. You have 12 minutes. Assign roles now: one writer, one who checks each calculation step, one who writes the flame-colour sentence, one who manages time.' Set a visible timer. After 12 minutes, say: 'Pens down. Mount your poster on the wall.' Hand each learner two sticky notes (one yellow = Star, one pink = Wish). Say: 'You will visit exactly two other posters. Spend 3 minutes at each. Write one star and one wish per poster and stick it on.' After the gallery walk, say: 'Return to your own poster. Read what your peers wrote. You have 2 minutes to discuss — do you agree with their wish? Would you change anything?' Cold-call one spokesperson from 3 groups to share the most interesting feedback they received. Ask the class: 'Which group's flame-colour sentence was clearest? Why?' Wait 15 seconds before cold-calling.	Gallery Walk with Star-and-Wish Peer Feedback — this strategy externalises learner thinking onto a public artefact, making reasoning visible and critiqueable. The structured feedback protocol (Star/Wish) ensures that peer review is specific and constructive rather than vague. The flame-colour sentence forces learners to explicitly reconnect the calculation back to the anchoring phenomenon, preventing the mathematics from becoming disconnected from the driving question.	Observable indicator: The flame-colour sentence on each poster must mention that isotopes share the same atomic number / electron arrangement / proton number, and therefore produce identical flame colours. Teacher marks this as the key conceptual indicator. If any group writes that flame colour depends on mass, teacher addresses this explicitly in Phase 4 whole-class discussion. Teacher also checks that all posters show the reasonableness check.

Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Isotopes and Atomic Mass Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Dalton's Atomic Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front of the room. The teacher facilitates a structured whole-class discussion in which learners propose updates to the board. Three columns are updated: (1) 'What we now KNOW' — learners suggest: 'RAM is a weighted average of isotope masses, weighted by their percentage abundance'; 'Chlorine's RAM ≈ 35.5 because Cl-35 is three times more abundant than Cl-37'; 'Carbon is almost entirely C-12, so its RAM ≈ 12.01.' (2) 'What DETERMINES flame colour' — learners confirm and the teacher anchors: 'Atomic NUMBER (protons) determines electron arrangement, which determines flame colour — NOT atomic mass.' (3) 'New questions we still have' — learners add open questions, e.g., 'If electrons determine colour, how exactly does an electron produce light?' (previewing Lesson 6). The teacher physically writes or sticks updated cards onto the DQB as learners dictate.	Point to the DQB and say: 'We have been adding to this board since Lesson 1. Today we can answer something that has been bothering us — how can chlorine have a mass of 35.5 if no atom actually has that mass?' Wait 10 seconds. Cold-call 2 learners. After responses, say: 'Exactly — it is a weighted average, like the average score in a class where not everyone scores the same.' Then ask: 'Now the big connection — sodium has two isotopes: Na-23 (99.0%) and Na-22 (0.97%). If you added Na-22 to a flame instead of Na-23, what colour would you see?' Wait 15 seconds. Cold-call 3 learners. Confirm: 'Yellow-orange — because both isotopes have 11 protons and identical electron arrangements. Mass does not change the flame colour.' Write on the DQB in large letters: 'FLAME COLOUR = ATOMIC NUMBER fingerprint, NOT atomic mass.' Ask: 'What new questions does today's lesson leave us with?' Accept 3–4 contributions and add them to the 'Still wondering' column. Point to the question about how electrons produce light and say: 'That is exactly where we are heading in Lesson 6.'	Collaborative DQB Anchoring — the DQB functions as a living, public record of the class's growing scientific model. By having learners dictate what should be written rather than the teacher filling it in, agency remains with learners. The deliberate act of writing 'FLAME COLOUR = ATOMIC NUMBER fingerprint' in large print creates a durable visual anchor that will remain on the board for the rest of the unit.	Observable indicator: At least 6 different learners contribute verbally to the DQB update without prompting. Teacher listens specifically for the correct causal chain: 'same atomic number → same electron arrangement → same flame colour.' If a learner conflates mass number with atomic number, teacher pauses and facilitates peer correction before moving on.
Model Building Phase 📹 **VIDEO:** Video: Isotopes and Atomic Mass Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:**	Learners individually update their running Atomic Model in their exercise book (begun in Lesson 1 and revised each lesson). Today's revision task: below their existing sketch of a chlorine atom (from Lesson 4), learners draw BOTH isotopes of chlorine side-by-side — Cl-35 and Cl-37 — labelling protons, neutrons, electrons, atomic number, and mass number for each. Then they add a 'Weighted Average' annotation	Say: 'Open your exercise book to your running atomic model. You have 8 minutes to add today's revision. Draw Cl-35 and Cl-37 side by side. Label everything. Add the RAM calculation as an annotation. Write the one-sentence flame-colour conclusion.' Set a visible 8-minute timer. Circulate and check that: (a) both isotopes show different neutron counts; (b) both show identical electron counts (17); (c) the	Cumulative Model Revision — by requiring learners to update a single running model rather than starting fresh each lesson, this strategy makes scientific knowledge-building feel cumulative and iterative — mirroring how real scientists refine models over time. The side-by-side isotope diagram with the RAM annotation bridges the calculation (mathematics) to the atomic structure (science concept), preventing procedural	Observable indicator: Teacher checks that the one-sentence flame-colour conclusion correctly attributes colour to electron arrangement (not mass). This is the key exit indicator for Lesson 5. Secondary indicator: the weighted average calculation is shown step-by-step (not just the answer). Teacher uses a simple 3-point stamp system: ✓✓ = both indicators met; ✓ = one indicator met; circle = learner needs to redo

Dalton's Atomic Theory (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	arrow pointing between the two isotopes, with the calculation RAM = (35 × 0.75) + (37 × 0.25) = 35.5 written explicitly. Finally, learners write one sentence beneath the model: 'Both isotopes have 17 protons and 17 electrons — so both would glow green in a flame test (copper-like), because flame colour depends on electron arrangement, not mass.' Teacher circulates and stamps or initials complete models.	weighted average calculation is written correctly; (d) the flame-colour sentence correctly attributes colour to electron arrangement / atomic number. After 8 minutes, ask 2 volunteers to hold up their exercise books and show their model to the class via the document camera or by holding them up. Ask the class: 'What did they do well? What is missing?' After feedback, say: 'This model is now yours — it is the most complete picture of an atom you have ever drawn. In Lesson 6, we will add one more layer: where exactly those 17 electrons sit inside the atom — in energy levels and orbitals.'	knowledge from becoming disconnected from conceptual understanding.	the sentence before Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Calculation accuracy — did the majority of learners correctly apply the weighted average formula, including the step of dividing percentage by 100? If more than one-third of learners made the 'simple average' error (35+37)/2, plan a 5-minute warm-up correction at the start of Lesson 6 using a new Kenya-context analogy (e.g., averaging tea-picker wages in Kericho weighted by number of workers). (2) Flame-colour connection — did learners independently and correctly write the flame-colour sentence in Phase 5 without teacher prompting? This is the critical conceptual bridge of the lesson. If fewer than half of learners made this connection independently, the DQB anchor phrase should be revisited verbally at the start of Lesson 6. (3) Gallery Walk quality — were the Star-and-Wish comments specific and substantive, or were they generic ('good work')? If comments were too vague, model a specific feedback sentence next time: 'Your reasonableness check was clear, but your flame-colour sentence did not mention electron arrangement.' (4) Pacing — the lesson is dense. If Phase 3 (poster + gallery walk) ran over time, consider having groups prepare the poster only (no gallery walk) and sharing via a whole-class display instead. Note which groups needed the most support with fractional abundance conversion and seat them near the teacher during Lesson 6 orbital work.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners received isotopic abundance data cards for chlorine (Cl-35: 75%, Cl-37: 25%) and carbon (C-12: 98.9%, C-13: 1.1%). They observed that the periodic table lists chlorine's atomic mass as 35.5 — a value between the two whole-number isotope masses — and investigated why this non-whole-number value appears.
What did I learn?	Learners learned that relative atomic mass (RAM) is calculated as a weighted average of isotope masses, using the formula RAM = Σ(isotope mass × fractional abundance). They calculated RAM for chlorine (≈ 35.5) and carbon (≈ 12.01), and verified their answers using a reasonableness check (RAM must fall between the lightest and heaviest isotope masses).
How does this explain the phenomenon?	Learners explained that because all isotopes of an element share the same atomic number and therefore the same electron arrangement, they produce identical colours in a flame test. Cl-35 and Cl-37 would both produce the same flame colour because they both have 17 protons and 17 electrons. Flame colour is a fingerprint of atomic number — not atomic mass. The Driving

	Question Board was updated to record: 'FLAME COLOUR = ATOMIC NUMBER fingerprint, NOT atomic mass.'

LESSON 6 (80 minutes): From Energy Levels to Orbitals: Introducing s and p Orbitals	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To transition learners from Bohr's shell model to the modern quantum orbital model by introducing the shapes, names, and electron capacities of s and p orbitals for the first 20 elements.
Knowledge	Learners explain the relationship between energy levels and orbitals; distinguish between s and p orbitals by shape and electron capacity; identify orbitals (1s, 2s, 2p, 3s, 3p) and their positions within principal energy levels.
Skills	Learners observe and interpret animations of orbital shapes (Digital Literacy); sort and match orbital labels to energy levels and capacities (card-sort); write basic s and p electron configurations for selected elements from the first 20; revise their cumulative atomic model.
Attitudes	Learners develop interest in the study of the structure of the atom; show integrity by appropriately using digital devices to watch animations on electron arrangement; advocate harmonious relationships while working in card-sort groups (Social Justice).
Key Inquiry Question	How are electrons arranged in an atom?
Purpose in Storyline	In Lessons 1–5 learners established that atoms have a nucleus with electrons in shells (Bohr model) and that electrons absorb and release specific amounts of energy — producing the flame colours seen at the Kenyan cooking hearth. Lesson 6 deepens this: the 'shells' are not uniform regions but contain orbitals of specific shapes and capacities. Understanding that sodium's single 3s electron and potassium's single 4s electron each release a unique quantum of energy explains why the two flames glow different colours. This lesson therefore reframes the flame-colour phenomenon in the language of the modern orbital model, setting up Lessons 7–9 in which learners write full s and p configurations for the first 20 elements.
Safety Notes	No hazardous chemicals are used in this lesson. If learners handle printed card-sort materials, remind them not to place small cards near their mouths. When using the ARES platform on tablets or laptops, learners must follow the school's Acceptable Use Policy for digital devices (Integrity — Values). Ensure tablet screens are kept clean and devices are returned to the charging station at the end of the lesson.

B. LESSON OVERVIEW

Learners begin by revisiting the anchoring phenomenon — the yellow-orange flash of sodium and the violet flash of potassium at the Kenyan cooking hearth — and are challenged to explain WHY two different elements in the same hot flame produce two different colours. After a brief predict phase, learners watch two short ARES-platform animations: (1) the limitation of Bohr shells and (2) the three-dimensional shapes of s and p orbitals. They then work in pairs on a structured card-sort activity, matching orbital names, principal energy level numbers, shapes, and maximum electron counts. The lesson closes with a whole-class DQB update and individual model-revision sketches. By the end, learners can articulate that the specific orbital an electron occupies determines the precise energy it releases — and therefore the precise colour of flame light produced.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners individually read a short	1. Display the Kisumu lab mystery	Think-Pair-Share with DQB	Observable indicator: Every

📹 **VIDEO:** [De Broglie wavelength] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Conservation of Energy in Chemical Reactions (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	'mystery card' displayed on the ARES platform: 'At a school lab in Kisumu, a student places a copper wire in a Bunsen flame — it glows green. Another student places a piece of steel wool — it glows orange-yellow. A third places a strip of magnesium ribbon nearby — it flares brilliant white. All three substances are in the SAME flame, at roughly the SAME temperature. Yet each gives a DIFFERENT colour.' Learners write a 2–3 sentence prediction in their exercise books answering: 'What is it about the INSIDE of a sodium atom versus a copper atom that makes them release different colours of light?' They then share predictions with a partner (Think-Pair-Share) before two or three pairs share with the class. Learners record their initial ideas on sticky notes to be placed on the Driving Question Board.	card on the ARES platform projector. Read it aloud once. Say: 'Before we learn anything new today, I want to know what YOU think. Write your prediction — even if you are unsure, your best scientific reasoning is what matters.' Allow 4 minutes of silent individual writing. 2. Say: 'Turn to your partner. You have 90 seconds each — share your prediction and listen carefully to theirs.' Circulate; listen for student language around 'energy', 'electrons', 'shells', or 'heat'. 3. After pair-share, cold-call EXACTLY 3 pairs using a random name stick. For each, ask a probing follow-up: 'You said the atom releases energy — but WHERE inside the atom does that energy come from?' Apply 12 seconds of WAIT TIME before accepting responses. 4. Acknowledge all predictions non-evaluatively: 'Thank you — these ideas are exactly the kind of thinking scientists had before the orbital model was developed.' 5. Instruct learners to write their prediction on a sticky note and place it in the 'Our Current Ideas' column of the DQB.	anchoring. This activates prior knowledge from Lessons 1–5 (Bohr shells, electron energy transitions) and creates cognitive dissonance — learners realise their current shell model cannot fully explain WHY different orbitals within a shell produce different energies. Recording predictions on the DQB makes thinking visible and creates accountability for revision later in the lesson.	learner has written at least 2 sentences in their exercise book before pair-share begins. Teacher scans books during circulation. Listen for whether learners reference 'electrons', 'energy levels', or 'shells' — this gauges retention from Lessons 2–5. Flag any learner who writes only 'because the atoms are different' for targeted questioning during the observe phase.
Observe Phase 📹 **VIDEO:** [De Broglie wavelength] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Conservation of Energy in Chemical	Learners watch two sequential animations on the ARES platform. Animation 1 (4 min): 'The Limit of Bohr Shells' — shows that within Shell 2 of sodium, electrons are NOT all in the same location or shape; the shell contains sub-regions. Animation 2 (6 min): 'Shapes of s and p Orbitals' — a 3D visualisation showing: the spherical shape of s orbitals (1s, 2s, 3s); the dumbbell/figure-8 shape of the three 2p orbitals (2px, 2py, 2pz) oriented along x, y, z	1. Before playing Animation 1, say: 'As you watch, I want you to look for one thing Bohr's model CANNOT explain that the new model can. Write it in your observation table under a column labelled: What Bohr missed.' Play Animation 1. Pause at 2:10 where the animation shows two electrons in Shell 2 at different energy sub-levels. Ask: 'Why are these two electrons at different energies if they are in the SAME shell?' Apply 10 seconds WAIT TIME. Cold-call	Structured Observation Table with Annotation. By requiring learners to fill a table during the animation (rather than passively watching), attention is directed to key features: shape, name, capacity. Annotating the previously drawn Bohr model creates a direct conceptual bridge — learners physically mark WITHIN Bohr's shells to show orbital sub-structure. This embeds NGSS Science and Engineering Practice 2 (Developing and Using Models)	Observable indicator: Learner observation tables contain correct entries for at least 1s, 2s, and 2p rows before the card sort begins. Teacher spot-checks 6 books during circulation (2 from front, 2 from middle, 2 from back rows). Specific misconception to flag: learners writing 'p orbital holds 2 electrons' — the three 2p orbitals together hold 6; each individual p orbital holds 2. Address immediately by drawing three separate dumbbell boxes on the

Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	axes; and the fact that each orbital holds a maximum of 2 electrons. As they watch, learners complete a structured observation table in their exercise books with three columns: 'Orbital Name | Shape | Maximum Electrons'. After the animations, learners spend 4 minutes discussing in pairs: 'How does this animation change or add to the Bohr model we drew in Lesson 3?' They annotate their previously drawn Bohr models with orbital labels.	2 students. 2. Say: 'Now we go deeper — watch for the SHAPES of the regions where electrons are found.' Play Animation 2. Pause at 1:45 (s orbital sphere) and ask: 'Describe the shape you see — use a Kenyan example if it helps.' (Accept: round like an orange, round like a maize grain.) Pause at 3:20 (p orbital dumbbell) and ask: 'How many of these dumbbell shapes are there, and how are they arranged?' Apply 12 seconds WAIT TIME. Cold-call 3 students. 3. After both animations, say: 'Complete your observation table now — you have 4 minutes. Then annotate your Bohr model from Lesson 3.' Circulate and check that learners are correctly labelling 1s (max 2e), 2s (max 2e), 2p (max 6e across 3 orbitals). 4. Pose bridging question to the whole class: 'A sodium atom has 11 electrons. If we fill these orbitals in order, which orbital holds the outermost electron?' Apply 15 seconds WAIT TIME. Do not confirm yet — say: 'Hold that thought for the card sort.'	and Practice 4 (Analysing and Interpreting Data).	board.
Explain Phase 📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Energy in Chemical Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs on a printed/digital card-sort activity. Each pair receives 20 cards (or views them on the ARES tablet) organised in four sets: SET A — Orbital Names (1s, 2s, 2px, 2py, 2pz, 3s, 3px, 3py, 3pz); SET B — Shapes (sphere, dumbbell ×3); SET C — Principal Energy Level (1, 2, 3); SET D — Maximum Electrons per orbital (2, 2, 2). Learners sort and physically arrange cards on their desk or drag-and-drop on ARES to form a complete matching table. After matching, each pair writes three 'because' statements, e.g.: '2p orbitals are at energy level 2	1. Distribute card-sort sets (or open on ARES tablet). Say: 'You have 8 minutes to sort all cards into a matching table. Work with your partner — use your observation table from the animations to help you.' Set a visible 8-minute countdown on the ARES platform. 2. Circulate purposefully — visit at least 6 pairs. When a pair correctly matches 2p with 'dumbbell' and 'energy level 2', affirm the reasoning: 'Good — now tell me WHY 2p is at level 2 and not level 1.' Apply 10 seconds WAIT TIME. 3. At the 8-minute mark, say: 'Now write your three BECAUSE	Card-Sort with 'Because Statement' Justification. The card sort operationalises NGSS Practice 6 (Constructing Explanations) by requiring learners to construct relational knowledge rather than memorise labels. The 'because statement' scaffold forces causal reasoning. Applying the sorted table immediately to Na and K directly reconnects to the anchoring phenomenon — learners see that the orbital holding the outermost electron is the proximate cause of the specific flame colour observed at the Kenyan cooking hearth. This also develops Learning to Learn	Observable indicator: Each pair produces a completed matching table AND three written 'because' statements before whole-class share. Teacher collects one 'because' statement set from 4 pairs (selected randomly) to check for causal language. Red-flag responses: pairs who write 'because that is the rule' rather than referencing energy levels — pull these pairs for a 2-minute re-teach using the orbital energy diagram on ARES. Specific check: learner can correctly state that Na's outermost electron is in 3s and K's outermost electron is in 4s.

	BECAUSE they are found within the second principal shell.' They then apply their sorted table to answer: 'Using your card sort, fill orbitals in order for Na (Z=11) and K (Z=19). Which orbital holds the outermost electron in each?' Pairs share answers with a neighbouring pair, then the class converges on a whole-group answer. The teacher uses the ARES projected display to reveal the canonical card-sort solution.	statements. These must use the words: orbital, energy level, and electrons.' Allow 4 minutes. 4. Say: 'Apply your sorted table: fill the orbitals for sodium — Z equals 11. Write the sequence step by step.' Cold-call 1 pair to present their Na sequence on the board. Ask the class: 'Do you agree or disagree? Show thumbs.' Address any errors — particularly the common error of skipping directly from 2s to 3s without filling 2p. 5. Reveal ARES solution display. Say: 'Look at sodium's outermost electron — it is in the 3s orbital. Look at potassium's outermost electron — it is in the 4s orbital. Now, think back to our phenomenon: these two electrons are in DIFFERENT orbitals at DIFFERENT energy levels. What does that mean for the colour of light they release?' Apply 15 seconds WAIT TIME. Cold-call 3 students. Record key phrases on the board: 'different orbital → different energy gap → different colour of light.'	competency as learners self-check their sorting against the ARES canonical display.	
Driving Question Board (DQB) Creation 📹 **VIDEO:** [De Broglie wavelength] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Conservation of Energy in Chemical Reactions (Flexbook)] Source: CK-12	Learners return to the class Driving Question Board. They retrieve their sticky notes from Phase 1 and individually re-read their original prediction. Each learner writes a 'Then vs Now' card: on one side they copy their original prediction; on the other side they write a revised explanation using orbital vocabulary. Specifically, learners respond to the DQB question strip: 'WHY do sodium and potassium produce different flame colours?' using the sentence frame: 'Sodium produces yellow-orange light because its outermost electron is in the ___ orbital. When heated, this electron absorbs energy and	1. Direct learners to the DQB. Say: 'Pick up the sticky note you wrote at the start of the lesson. Read it carefully. How has your thinking changed?' Allow 2 minutes of silent reflection. 2. Distribute 'Then vs Now' cards (index cards or printed half-sheets). Say: 'On side A, copy your original prediction word for word. On side B, write your revised explanation using at least THREE of these words: orbital, energy level, 3s, 4s, energy gap, colour, electron.' Display the sentence frame on ARES. Allow 4 minutes. 3. Cold-call 3 volunteers (use random name sticks) to read their side B aloud. After each, ask the class: 'What orbital vocabulary	Then vs Now Reflective Card with DQB Anchoring. This metacognitive strategy (Learning to Learn competency) makes conceptual change visible — learners physically compare their pre-instruction prediction with their post-instruction explanation. The sentence frame scaffolds use of correct orbital vocabulary without removing cognitive demand. Adding new questions to the DQB models authentic scientific practice: answering one question always generates new ones (NGSS Practice 1 — Asking Questions).	Observable indicator: Side B of every learner's 'Then vs Now' card contains at least one specific orbital name (e.g., 3s or 4s) AND a causal link to flame colour. Teacher collects all cards at the end of the lesson for written feedback. Flag: learners who still write 'electrons move to a higher shell' without specifying the orbital — these learners need targeted support at the start of Lesson 7. Count of new DQB questions: aim for at least 5 genuine new questions posted, indicating productive curiosity.

🔍 Search ARES for similar readings	jumps to a higher orbital. When it falls back, it releases ___J of energy as ___ (colour) light. Potassium's outermost electron is in the ___ orbital, so it releases a different amount of energy and produces ___ (colour) light.' Three volunteers read their completed cards aloud. The class votes on the most complete scientific explanation using the ARES response tool. New questions generated by the lesson are added to the 'New Questions' column of the DQB.	did they use correctly? What could make this explanation stronger?' Apply 10 seconds WAIT TIME between each reading. 4. Say: 'What NEW questions has today's lesson raised for you? Write one on a fresh sticky note and add it to the New Questions column of the DQB.' Scan the new questions; read two aloud: 'Great — these are exactly the questions we will investigate in Lessons 7 and 8.' 5. Summarise: 'Today we moved from Bohr's shells to orbitals. Sodium's yellow flame and potassium's violet flame are not random — they are a direct consequence of WHICH orbital holds the outermost electron and how much energy that orbital's electron releases as light.'		
Model Building Phase 📹 **VIDEO:** De Broglie wavelength Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Conservation of Energy in Chemical Reactions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative atomic model — a layered diagram they have been building since Lesson 1 (starting with Dalton's solid sphere, adding Rutherford's nucleus in Lesson 2, adding Bohr's shells in Lesson 3, adding the nucleus composition in Lessons 4–5). Today, learners ADD Layer 6: they redraw Shell 2 of their model to show the 2s sub-region (small sphere) and the three 2p sub-regions (three dumbbells along x, y, z axes) inside the shell boundary. They label: 1s (2e), 2s (2e), 2p (6e). They then draw a SODIUM atom model showing: $1s^2$ $2s^2$ $2p^6$ $3s^1$ — with the lone 3s electron highlighted in a different colour and annotated: 'This electron releases yellow-orange light (589 nm) when it falls back from an excited orbital — this is what the cook in the Rift Valley sees in the fire!' Learners write a one-sentence caption connecting	1. Say: 'Open your cumulative model. You have been building this since Lesson 1. Today we add the most important layer yet — the orbital sub-structure inside Bohr's shells.' Display the Layer 6 instruction diagram on ARES (showing 2s sphere nested inside Shell 2, and three 2p dumbbells). Allow 1 minute for learners to orient themselves. 2. Say: 'You have 8 minutes to add Layer 6 to your model AND draw sodium's full orbital model labelled $1s^2$ $2s^2$ $2p^6$ $3s^1$. Use a different colour for the 3s electron.' Circulate and provide corrective feedback. Watch specifically for: (a) learners drawing p orbitals as circles rather than dumbbells — correct immediately with a quick sketch; (b) learners placing 3 electrons in one 2p orbital rather than 2 — refer them to their card-sort table. 3. At the 8-minute mark, say: 'Write your one-sentence caption	Cumulative Model Revision with Phenomenon Caption. The iterative model-building strategy embeds NGSS Practice 2 (Developing and Using Models) and ensures that each new concept is integrated into a growing, coherent representation rather than learned in isolation. The mandatory phenomenon caption prevents the model from becoming an abstract exercise — learners must always trace their diagram back to the observable flame colour at the Kenyan cooking hearth. Peer star-and-step review develops self-regulation and collaborative critique skills (Learning to Learn; Social Justice values).	Observable indicator: Learner's Layer 6 diagram correctly shows s orbital as sphere and p orbitals as dumbbells, with labels 1s(2e), 2s(2e), 2p(6e). The sodium model shows $1s^2$ $2s^2$ $2p^6$ $3s^1$ with the 3s electron distinctly marked. The phenomenon caption explicitly names the flame colour AND links it to the 3s orbital. Collect 6 models (random sample) to assess before Lesson 7. Use a 3-point rubric: 1 = shape and label errors present; 2 = correct shapes and labels but no phenomenon link; 3 = correct shapes, labels, and explicit phenomenon link in caption.

	their model to the phenomenon. Pairs compare models and give each other one 'star' (what is correct) and one 'step' (what to improve).	connecting your sodium model to the Rift Valley cooking hearth phenomenon.' Allow 2 minutes. Cold-call 2 learners to read their captions. 4. Say: 'Now, peer-review with your partner. Give one star and one step.' Allow 2 minutes. 5. Close the phase by saying: 'By Lesson 9, your model will show the full s and p electron arrangement for all 20 elements. Today's sodium model is the foundation. In our next lesson we will practise writing s and p configurations for more elements — and you will begin to see the pattern that builds the periodic table.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. CONCEPTUAL CHANGE: Did learners' 'Then vs Now' cards show genuine movement from Bohr shell language to orbital language? How many learners (out of your class total) correctly named 3s as sodium's outermost orbital on their DQB card? If fewer than 70% did, plan a 5-minute orbital-naming warm-up for Lesson 7.

2. DIGITAL LITERACY: Did all learners actively complete their observation tables during the animations, or did some passively watch? Consider whether the ARES pause points were at the right moments — adjust pause timing in the platform settings if needed.

3. CARD SORT DIFFERENTIATION: Which pairs struggled most with the card sort? Were errors in the 'shape' set (confusing s and p shapes) or in the 'energy level' set (misassigning orbitals to principal levels)? Prepare a simplified 2-column version (orbital name ↔ energy level only) as a scaffold for Lesson 7 for those pairs.

4. PHENOMENON CONNECTION: Did learners spontaneously use the cooking hearth / Kisumu lab phenomenon in their model captions, or did it feel forced? If the phenomenon link felt mechanical rather than meaningful, consider opening Lesson 7 with a 2-minute re-viewing of the flame colour video clip from Lesson 1 to re-anchor the storyline.

5. INCLUSION AND GENDER: Were contributions during cold-calling balanced across gender and across learners from different sub-counties? Review your name-stick records and ensure underrepresented voices are intentionally included in Lesson 7 cold-calls.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed two ARES animations: (1) the limitation of Bohr's electron shells in explaining sub-level energy differences, and (2) the three-dimensional shapes of s orbitals (spherical) and p orbitals (dumbbell/figure-8 along x, y, z axes). Students also observed that sodium (Z=11) has its outermost electron in the 3s orbital and potassium (Z=19) has its outermost electron in the 4s orbital — two different orbitals at two different energy levels.
What did I learn?	Students learned that within each principal energy level (shell), electrons occupy sub-regions called orbitals; that s orbitals are spherical and hold a maximum of 2 electrons; that there are three p orbitals (px, py, pz) per principal level, each holding 2 electrons (6 electrons total in p sub-level); that orbitals fill in order 1s → 2s → 2p → 3s → 3p; and that the specific orbital holding an atom's

	outermost electron determines the precise energy gap — and therefore the precise colour of light — released when that electron returns from an excited state.
How does this explain the phenomenon?	Students explained the anchoring phenomenon more precisely than before: the sodium atom's lone outermost electron occupies the 3s orbital; when heated in a flame, it absorbs energy and jumps to a higher orbital, then releases exactly 3.37×10^{-19} J as yellow-orange light (589 nm) when it falls back. Potassium's outermost electron in the 4s orbital has a different energy gap, so it releases a different quantum of energy, producing violet light. This is why different substances — salt at a Kenyan cooking hearth, or copper wire in the Kisumu school lab — always produce their own unique, repeatable flame colour: it is a direct fingerprint of orbital structure.

LESSON 7 (80 minutes): Filling the Orbitals: Aufbau, Pauli, and Hund's Rules	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to discover and apply the three rules governing how electrons fill orbitals — the Aufbau principle, the Pauli exclusion principle, and Hund's rule — and to write orbital (box) diagrams and s/p notation for elements 1–10.
Knowledge	Learners will know that: (i) electrons fill orbitals in order of increasing energy (Aufbau principle); (ii) each orbital holds a maximum of two electrons with opposite spins (Pauli exclusion principle); (iii) electrons occupy each orbital of equal energy singly before pairing (Hund's rule); and (iv) the specific orbital an electron occupies determines the exact energy it releases as light, explaining why sodium always emits the same yellow colour.
Skills	Learners will be able to: draw orbital box diagrams for elements 1–10; write electron configurations in s and p notation; apply all three filling rules correctly; and identify errors in given orbital diagrams.
Attitudes	Learners develop interest in the study of the structure of the atom and show integrity by self-disciplining their use of digital simulation tools to explore electron arrangements.
Key Inquiry Question	How are electrons arranged in an atom?
Purpose in Storyline	In Lessons 1–6, learners established that atoms have a nucleus, that electrons occupy energy levels/shells, and that each sublevel (s, p) contains orbitals of defined shape. Lesson 7 now zooms into those orbitals and asks: in exactly what ORDER and PATTERN do electrons fill them? Discovering the three rules provides the mechanism that will explain, in Lesson 8, why the sodium electron that absorbs flame energy always starts in the 3s orbital and always emits exactly 589 nm yellow light when it falls back — the same yellow flash the Rift Valley cook saw when salt landed in the fire.
Safety Notes	No hazardous chemicals are used in this lesson. If a Bunsen burner is used for a brief recall demonstration, follow standard lab safety: tie back hair, no loose clothing, use tongs, keep flammable materials away from the flame. The main activity uses paper-based 'electron seat' cards — no safety concerns.

B. LESSON OVERVIEW

Learners begin by predicting how electrons might arrange themselves in orbitals, drawing on their knowledge of energy levels and sublevels from Lessons 5–6. They then carry out a structured 'Electron Seating' guided-inquiry activity using orbital box diagrams, discovering the Aufbau principle, Pauli exclusion principle, and Hund's rule through evidence and discussion. A short PhET simulation segment reinforces the rules visually. Learners practise writing orbital diagrams and s/p notation for elements 1–10. The lesson closes by explicitly linking the specific orbital occupancy of sodium's outermost electron to the yellow flame colour seen at the Kenyan cooking hearth — advancing the class's answer to the driving question on the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners individually examine a printed 'Orbital Seating Chart' — a	Display the projected 'Electron Seating Chart' and the orbital	Elicitation of Prior Knowledge via Predict-Then-Compare: surfacing	Circulate and note: (1) Do learners respect the 1s < 2s < 2p energy

📹 **VIDEO:** [Electron configurations for the first period](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electron Cloud Atomic Model (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	row of empty boxes labelled 1s, 2s, 2p, 2p, 2p, 3s — and are asked to place tokens (small torn paper squares) representing electrons for carbon (6 electrons) however they think makes sense. They write a 2–3 sentence justification on a sticky note explaining their choices before any instruction is given. Pairs then share predictions and note disagreements. The class is reminded of the anchoring phenomenon: 'You already know sodium gives yellow light and potassium gives violet. Today we find out WHY — by discovering the exact rules that decide which orbital each electron sits in.'	boxes on the board. Say exactly: 'Before I teach you anything new, I want you to be a scientist. Here is carbon — 6 electrons. Using what you already know about energy levels and orbitals, place the electrons in these boxes in the way YOU think is most logical. There is no wrong answer yet — this is your prediction.' Allow 4 minutes of silent individual work. Then say: 'Turn to your partner. Compare your diagrams. Where do you agree? Where do you disagree? Be ready to defend your choice.' Allow 3 minutes of pair talk. Cold-call 3 pairs (select one high-confidence, one uncertain, one who placed all electrons in 1s): 'Achieng, walk us through your diagram — why did you place electrons there?' WAIT 12 seconds after each response before probing. Record 3 different class predictions visibly on the board and label them Prediction A, B, C. Say: 'We will return to these at the end of the lesson and see which one nature chose — and why.'	and recording multiple competing predictions creates cognitive dissonance that motivates the evidence-gathering phase. Publicly posting Predictions A, B, C gives learners a concrete claim to test, making the subsequent activity purposeful rather than procedural.	ordering from Lesson 6? (2) Do any learners spontaneously apply a pairing rule? (3) Can learners articulate any justification at all? Flag learners who place all 6 electrons in 1s — they need targeted re-teaching of energy levels before Phase 2. Look for sticky-note justifications that reference 'lower energy first' as evidence of readiness.
Observe Phase 📹 **VIDEO:** [Electron configurations for the first period](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electron Cloud Atomic Model (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners carry out the 'Electron Seat' structured discovery activity in groups of 4. Each group receives: a large A3 'Orbital Energy Ladder' sheet (boxes stacked 1s at bottom, then 2s, then three 2p boxes side-by-side, then 3s at top); a set of 20 small arrow-cards (red arrows pointing UP = spin-up; blue arrows pointing DOWN = spin-down); and an Element Strip showing elements H through Ne with atomic numbers 1–10. Groups follow a step-by-step instruction card: Step 1 — Place electrons for H (1 electron). Step 2 — Place electrons for He (2 electrons). Step 3 — Place	Distribute materials and read aloud the three riddle-rules together as a class. Say: 'These three rules are the actual rules nature uses. Your job is to discover their meaning by doing — not by copying notes.' Circulate continuously. At the 5-minute mark, check every group's He diagram: if any group places both electrons spin-parallel, crouch to eye level and ask: 'Your two arrows in 1s — can you show me something in Rule B that would allow that arrangement?' WAIT 15 seconds. At the 10-minute mark, spotlight a group that has correctly filled nitrogen (N: ↑ in each 2p box, none doubled): 'Freeze —	Structured Discovery / Riddle-Rule Induction: by encoding the three principles as observable constraints on a physical placement task rather than as definitions to memorise, learners induce the rules from evidence — mirroring how Pauli and Hund derived them from spectroscopic data. The Arrow-Card manipulative externalises the abstract concept of spin, making the Pauli exclusion principle tangible. The PhET verification step introduces digital literacy as a cross-checking tool, reinforcing KICD's digital literacy competency.	Collect group A3 sheets at the end of Phase 2. Scan specifically for: (1) Correct Hund's rule application at carbon (↑ in 2s, ↑ in one 2p, ↑ in second 2p — NOT paired in one 2p box). (2) Opposite-spin pairing in He and all subsequent filled orbitals. (3) Aufbau order respected — no electrons in 2p before 2s is filled. Groups with errors in carbon or nitrogen are flagged for targeted reteaching in Phase 3.

	electrons for Li (3 electrons). Continue through to Ne (10 electrons). Groups must follow three 'Seating Rules' printed on the card — but the rules are written as RIDDLES, not as named principles: Rule A: 'Always fill the lowest available seat first.' Rule B: 'Each seat holds at most 2 arrows, and the 2 arrows in one seat must point in OPPOSITE directions.' Rule C: 'When seats are at the same height (same energy), put one arrow in each seat before you double up.' After completing the ladder for all 10 elements, groups examine the PhET 'Build an Atom' simulation (if devices are available) or a teacher-projected walkthrough to verify their placements for C, N, and O. Learners record any corrections and annotate why they made an error.	everyone look at Group 3's nitrogen. What do you notice about the three 2p electrons?' Cold-call 2 learners from other groups to describe what they see. At 15 minutes, ask all groups: 'Which rule did you find hardest to apply, and why?' Collect verbal responses from 3 groups. If devices are available, project PhET for 3 minutes and narrate: 'Watch the simulation add electrons one at a time. Call out which rule it is following at each step.'		
Explain Phase 📹 **VIDEO:** Electron configurations for the first period Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher formally names the three rules discovered in Phase 2, connecting each rule-name to the riddle-rule learners used. Learners convert their arrow-box diagrams into written s/p notation for all elements 1–10, recording configurations in their exercise books. They then tackle a 'Spot the Mistake' challenge: four printed orbital diagrams are shown (for B, C, O, and F), two of which contain deliberate errors (a Hund's rule violation and a Pauli violation). Learners individually identify and correct the errors, writing one sentence explaining which rule was broken and why. The lesson then pivots back to the phenomenon: learners are shown a diagram of sodium (Na, element 11) and asked to extend their rules to write Na's configuration:	Begin by projecting a summary table with three columns: Riddle Rule | Scientist's Name | Formal Statement. Read each riddle rule aloud, then say: 'You discovered this rule yourselves. Scientists gave it a name.' Formally introduce: Aufbau principle (Aufbau means 'building up' in German — a detail that sparks curiosity), Pauli exclusion principle, Hund's rule. Say: 'Write these in your book — not as new facts, but as names for what you already discovered.' For 'Spot the Mistake', project each diagram for 90 seconds. Say: 'Work alone. Circle the error. Name the rule it breaks. Write one sentence.' WAIT until at least 80% of learners are writing before calling on anyone. Cold-call 4 different learners (one per diagram) using a name-stick jar.	Concept Naming After Discovery (Labelling-Last Strategy): formal vocabulary is introduced AFTER learners have operationalised the concept, preventing the illusion of understanding from mere memorisation. The 'Spot the Mistake' task promotes deeper processing through error analysis — a higher-order thinking move. The explicit sodium pivot is a Phenomenon Connection Moment — linking abstract orbital rules directly to the observable yellow flame, sustaining the driving question's relevance.	Collect 'Spot the Mistake' individual sheets. Correct identification of both errors (Hund's violation + Pauli violation) with accurate rule citation indicates mastery of Phase 2 content. Observe whether learners can extend Aufbau correctly to Na (element 11) without prompting — this is the bridge to Lesson 8. Use a quick show-of-hands: 'Who can now explain, in one sentence, why sodium always gives yellow?' A target of 70%+ of learners able to articulate the electron-jump explanation signals readiness to proceed.

	$1s^2 2s^2 2p^6 3s^1$. The teacher draws attention to the lone $3s^1$ electron and says: 'This single electron in 3s is the one that absorbs energy from the flame and jumps to a higher orbital. When it falls back to 3s, it releases light. The exact energy gap between those orbitals determines the colour. Because sodium always has this same electron arrangement, it always releases the same energy — always yellow.' Learners annotate their Na diagram with an arrow showing the electron's jump and fall, and label it: 'Source of the yellow flame colour.'	After each response, ask the class: 'Thumbs up if you agree, thumbs sideways if you're unsure.' Address any thumbs-sideways. For the sodium pivot, say: 'We've been building to this moment. Sodium — element 11. Use your three rules and extend your diagram past neon. What configuration do you get?' Give 3 minutes. Cold-call one learner to write on the board. Say: 'Now look at that $3s^1$ electron. That electron is the reason the Rift Valley cook saw yellow light. Let me show you why.' Draw the energy-level jump diagram on the board, narrating each step.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Electron configurations for the first period] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electron Cloud Atomic Model (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the GREEN note, they write one thing they can now explain on the Driving Question Board that they could not explain before this lesson (e.g., 'I can now explain why sodium's electron always falls back with the same energy, giving the same yellow colour'). On the YELLOW note, they write one question that this lesson has raised or left unanswered (e.g., 'What happens when an electron absorbs so much energy it leaves the atom entirely?'). Learners post both notes on the class DQB under the columns 'We Can Now Explain' and 'Still Wondering'. The class reads three green and three yellow notes aloud. The teacher explicitly maps lesson progress: 'We have now answered HOW electrons fill orbitals. Our next step is to ask WHAT HAPPENS when the flame gives those electrons extra energy — and that will complete our explanation of the phenomenon.'	Hand out sticky notes and say: 'You have 3 minutes — green note first, then yellow.' Circulate to ensure every learner writes substantively (not just 'I learned the Aufbau principle' — push for phenomenon-linked statements). After posting, select three green notes that explicitly mention the sodium electron or the flame phenomenon and read them aloud: 'Listen to how [name] has connected today's rules to what we saw in the flame.' For yellow notes, select one that anticipates Lesson 8 content (electron excitation/emission) and say: 'This question — [read it] — is exactly where we are going next lesson. Hold that thought.' Update the DQB master tracker on the projector, moving 'How do electrons fill orbitals?' from the 'Investigating' column to the 'Answered' column.	Driving Question Board Curation: the DQB functions as a collective working memory for the class, making conceptual progress visible and sequential. Distinguishing 'We Can Now Explain' from 'Still Wondering' teaches learners that science is an ongoing process of narrowing uncertainty — a metacognitive habit central to KICD's 'Learning to Learn' competency. The teacher's explicit connection to Lesson 8 maintains narrative momentum in the storyline.	Read all green sticky notes after class. Evidence of mastery: green notes that correctly articulate the orbital-to-colour connection (e.g., 'the specific orbital determines the energy of light released'). Evidence of a gap: green notes that say only 'electrons fill lowest energy first' without connecting to the phenomenon. Yellow notes that reveal persistent misconceptions (e.g., 'why don't all elements glow the same colour since electrons all follow the same rules?') should be addressed at the start of Lesson 8.

Model Building Phase 📹 **VIDEO:** [Electron configurations for the first period](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electron Cloud Atomic Model (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their running 'Atomic Model' diagram — a cumulative drawing they have been revising since Lesson 1. For this lesson's revision, they add a new layer inside the atom: orbital boxes drawn to scale on the energy-level diagram, with arrows showing actual electron placements for the element they have been tracking all unit (each learner was assigned a different element from H to Ca at Lesson 1). Learners who were assigned sodium or potassium draw a special annotation: a curved arrow showing the flame-excited electron jumping from its ground-state orbital to a higher orbital and emitting a photon (drawn as a zigzag arrow) as it falls back, labelled with the colour of light produced. All learners write a 2-sentence model statement: 'In my element [X], the outermost electron occupies the [orbital]. If this electron absorbs energy from a flame, it would jump to [higher orbital] and release [predicted colour or 'unknown colour'] light when it returns.' The cumulative model is signed and dated to mark its revision.	Say: 'Open your running Atomic Model. Today you are adding the most detailed layer yet — the actual orbital filling pattern for your element.' Project a completed model for helium as an example, showing orbital boxes inside the electron-cloud diagram. Circulate and provide targeted prompts: For learners struggling with elements beyond Ne, say: 'Your element has more than 10 electrons — use your Aufbau order list and keep going past 2p. What comes after 2p?' WAIT 12 seconds before assisting. For learners assigned Na or K, say: 'You have the star elements today — the ones linked directly to the phenomenon. Make your arrow dramatic. Label the colour.' In the final 3 minutes, ask 2 learners (one assigned Na, one assigned K) to hold up their model and explain their annotation to the class. Close by saying: 'Every time you revise this model, it becomes more powerful. You started with a simple nucleus and shell in Lesson 1. Now you have a model that can explain why fire is yellow or violet. That is the power of understanding electron arrangement.'	Cumulative Model Revision: the running model functions as a personal theory-building artefact — each lesson's additions make prior knowledge visible and require learners to integrate new understanding with old. Assigning different elements distributes cognitive work and creates a classroom resource where peers can consult each other's models. The sodium/potassium annotation creates a direct, student-authored link between orbital theory and the anchoring phenomenon, making the explanation personally constructed rather than teacher-transmitted — fulfilling KICD's Creativity and Imagination competency.	Spot-check 6–8 models during circulation, looking for: (1) Correct orbital box diagram for the assigned element (Aufbau, Pauli, Hund all respected). (2) Written model statement that names the specific orbital of the outermost electron. (3) For Na/K learners: a correctly drawn excitation-emission arrow with a colour label. Models that omit the orbital boxes or revert to simple shell diagrams indicate the learner has not yet integrated sub-strand content on orbitals — flag for one-to-one support at lesson start of Lesson 8.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes:

1. PREDICTION QUALITY: Were learners' Phase 1 predictions informed by prior lessons (energy levels, sublevels)? If most predictions ignored energy ordering, Lessons 5–6 content may need revisiting before Lesson 8.

2. HUND'S RULE DIFFICULTY: Did most groups correctly apply Hund's rule at carbon and nitrogen, or did they default to pairing electrons immediately? Hund's rule is consistently the hardest of the three to internalise — if more than 30% of groups made errors, plan a 5-minute warm-up activity at the start of Lesson 8 using a 'seat-filling analogy' (e.g., passengers on a matatu filling window seats before sharing).

3. PHENOMENON CONNECTION: Did learners articulate, in their own words, why sodium's $3s^1$ electron is responsible for yellow light? Could they distinguish this from the general statement 'electrons release light when they fall'? The specificity of the orbital-to-colour link is the conceptual target for this lesson.

4. DQB QUALITY: Were green sticky notes phenomenon-linked, or generic? If generic, consider modelling a strong green-note statement at the start of future DQB

updates.

5. PACING: The 5-phase structure is ambitious for 80 minutes. If Phase 2 runs long (groups taking more than 20 minutes on the Electron Seating activity), abbreviate Phase 5 model-building to a home assignment — learners complete orbital annotations at home and bring models to Lesson 8 for peer review.

6. EQUITY CHECK: Were learners who were assigned elements beyond neon (Na–Ca) adequately supported in the model-building phase? Consider pairing them with a peer who has confidently mastered elements 1–10 before assigning elements 11+.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed the 'Electron Seating Chart' manipulative activity in which arrow-cards (representing electrons with defined spins) were placed into orbital boxes for elements hydrogen through neon, revealing patterns in how electrons fill orbitals. Learners also observed a PhET simulation (or teacher-projected walkthrough) showing electrons being added to an atom one at a time, visually confirming the filling order.
What did I learn?	Learners learned the three rules that govern electron filling: (1) the Aufbau principle — electrons fill the lowest available energy orbital first (1s before 2s before 2p before 3s); (2) the Pauli exclusion principle — each orbital holds at most 2 electrons, and they must have opposite spins (one arrow up, one arrow down); (3) Hund's rule — when orbitals of equal energy are available (e.g., the three 2p orbitals), one electron enters each before any orbital is doubly occupied. Learners also learned to write electron configurations in s/p notation for elements 1–10 and extended this to sodium ($1s^2 2s^2 2p^6 3s^1$).
How does this explain the phenomenon?	These three rules explain why sodium always produces the same yellow-orange colour in a flame: sodium's electron configuration always places its outermost electron in the 3s orbital (as determined by the Aufbau principle). When the flame provides energy, that specific 3s electron — and no other — absorbs it and jumps to a higher orbital. When it falls back to 3s, it releases exactly the energy corresponding to that orbital gap, which is the energy of yellow-orange light (589 nm). Because every sodium atom has this same orbital arrangement (enforced by the three filling rules), every sodium atom releases the same colour — explaining the consistent yellow flash seen when salt falls into the cooking fire at the Rift Valley homestead.

LESSON 8 (80 minutes): Electron Configuration for Elements 1–20 (s and p Notation)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to systematically write the full s and p electron configurations for elements 1–20, identify group patterns in outer electron configurations, and connect those patterns to the repeating flame colours observed in the anchoring phenomenon.
Knowledge	Learners will know: (a) the correct order of filling s and p orbitals for elements hydrogen (Z=1) through calcium (Z=20); (b) that elements in the same group share the same type and number of outermost electrons; (c) that small differences in energy levels across elements in the same group produce distinct but related flame colours.
Skills	Learners will be able to: (a) write the full s and p electron configuration for any of the first 20 elements using correct notation (e.g., $1s^2\ 2s^2\ 2p^6\ 3s^1$ for sodium); (b) identify the outermost subshell and relate it to group membership; (c) peer-teach an assigned element's configuration using correct chemical language.
Attitudes	Learners will: (a) develop interest in the study of the structure of the atom by appreciating how systematic electron filling rules reveal hidden order in matter; (b) demonstrate integrity by self-correcting their own written configurations during peer review; (c) advocate harmonious group relationships (social justice) during the peer-teaching exercise.
Key Inquiry Question	How are electrons arranged in an atom?
Purpose in Storyline	This is Lesson 8 of 9 in the evidence-gathering sequence. By now students can describe energy levels, orbitals, and the Aufbau/Pauli/Hund filling rules. In this lesson they apply all prior knowledge to write and read complete s and p configurations for all first 20 elements — moving from isolated rules to a systematic map of the atom. The peer-teaching exercise surfaces the Learning to Learn competency, and the closing Think-Pair-Share explicitly reconnects configurations to the anchoring phenomenon: why do Group 1 metals (Li, Na, K) all produce warm-toned flame colours, yet each colour is uniquely their own? Students leave ready for Lesson 9, where they will use these configurations to explain periodic trends and finalise their explanatory models of the Kenyan cooking-hearth phenomenon.
Safety Notes	No hazardous chemicals or open flames in this lesson. If any printed materials are used with a Bunsen burner demonstration for recall, standard PPE (lab coat, safety goggles) applies. Ensure adequate ventilation in the classroom. Students with colour-vision deficiency should be paired with a sighted partner during any colour-reference discussions; use labelled colour cards as an accessibility support.

B. LESSON OVERVIEW

Learners work in pairs to systematically write the full s and p electron configurations for all 20 elements from hydrogen (H) to calcium (Ca), using Aufbau order practised in Lesson 7. A structured peer-teaching exercise then has each pair explain their assigned element's configuration to another pair, developing the Learning to Learn competency. The lesson centres on a Think-Pair-Share that reconnects configurations to the anchoring phenomenon — the Kenyan cooking-hearth flame colours — by asking why Group 1 metals (Li, Na, K) produce similar but distinct colours. Students update the Driving Question Board (DQB) with the key insight: same outermost orbital type → similar chemical behaviour and similar flame-colour family; slight difference in energy level → unique emission wavelength → unique colour. The lesson closes with a cumulative model revision in which students annotate their atomic diagrams with full s and p notation for sodium and potassium specifically.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Electron configurations for the first period] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electron Cloud Atomic Model (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a half-sheet 'Prediction Card' listing six elements: H, C, Na, Mg, Cl, Ca. Without notes, they predict the full s and p electron configuration for each element and write a one-sentence justification. They also answer: 'Sodium and potassium are both in Group 1 and both produce warm flame colours. Predict — are their electron configurations completely the same, mostly similar, or completely different? Give a reason.' Predictions are not collected; they are folded and kept for comparison at the end of the lesson.	Distribute Prediction Cards face-down. Say: 'Do not flip the card until I say go. This is your thinking, no notes, no partner — just what you know right now.' After 30 seconds of silence, say 'Go.' Allow 5 minutes of silent individual work. Circulate silently; do NOT correct errors — note 2–3 interesting (correct or incorrect) predictions to reference later. After 5 minutes, say: 'Fold your card in half and write your name on it. We will come back to these at the end of class.' Use this scanning time to identify which elements students find hardest — typically Cl (2,8,7 → $1s^22s^22p^63s^23p^5$) and Ca (2,8,8,2 → $1s^22s^22p^63s^23p^64s^2$) — and plan targeted cold-calls for Phase 3.	Activating Prior Knowledge via Prediction Cards — By committing predictions to paper before instruction, students surface their current mental models of orbital filling. This cognitive 'stake in the ground' increases attention during the Observe phase because learners are motivated to check their own thinking. It also gives the teacher real-time diagnostic data on misconceptions (e.g., students who write 3d before 4s for Ca).	Observable indicator: Every student has written at least one full configuration attempt and a sentence justification for the Group 1 comparison question within 5 minutes. Teacher scans for the most common error pattern (e.g., stopping at $2p^6$ for sodium, omitting $3s^1$) to address explicitly in Phase 3. No grade is assigned — the card is a personal learning tool.
Observe Phase 📹 **VIDEO:** [Electron configurations for the first period] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electron Cloud Atomic Model (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Part A — Paired Systematic Configuration Practice (25 min): Pairs receive a laminated 'Configuration Table' worksheet listing all 20 elements with atomic number, a blank 'Configuration' column, and a blank 'Outermost Subshell' column. Using the Aufbau diagonal diagram from Lesson 7 (posted on the board), pairs collaboratively write the full s and p configuration for every element H through Ca, then circle the outermost subshell. They then colour-code: red for s-block outer electrons, blue for p-block outer electrons. Part B — Colour-Coded Group Pattern Discovery (10 min): Pairs shade the cells for all Group 1 elements (H, Li, Na, K) in one colour, Group 2 (Be, Mg, Ca) in a second, and the halogens (F, Cl) in a third. They answer two guided questions on the worksheet: (1)	Post the Aufbau diagonal diagram prominently. Say: 'You and your partner are going to build the complete electron configuration map for the first 20 elements — this is the most important table you will make all term. Use the Aufbau diagram as your guide. Fill every row before moving to the next.' After 8 minutes, pause the class. Cold-call 3 pairs (choose students who rarely volunteer): 'Pair at the back window — what did you write for oxygen?' Allow 10 seconds WAIT TIME before accepting the answer. Affirm correct responses: 'Exactly — $1s^22s^22p^4$. Notice oxygen is still in the 2p subshell.' For Part B, say: 'Now look only at the Group 1 column — Li, Na, K. What pattern do you see in the outermost subshell? Whisper your answer to your partner.' After 15 seconds: 'Group near the door —	Colour-Coded Pattern Mapping — The dual colour-coding (red for s-outer, blue for p-outer) makes the block structure of the periodic table visible on the configuration table itself. When students shade Group 1 elements and immediately see that each has a lone 'ns^1' outer configuration, the pattern is perceived visually before it is stated verbally. This multimodal encoding (writing + colour + pattern recognition) deepens retention and directly prepares the Think-Pair-Share in Phase 3 about flame colours. Peer cross-checking with a green pen operationalises the 'Learning to Learn' competency by making error-identification a valued, non-punitive act.	Observable indicators: (1) By the end of Part A, each pair has a fully completed 20-row table with outermost subshell circled for every element. (2) In Part B, students correctly identify that all Group 1 elements end in ns^1. (3) In Part C, fewer than 3 uncorrected errors remain per table after peer review. Teacher notes which elements still have errors after cross-checking — these become the focus of targeted explanation in Phase 3.

	'What do all Group 1 elements have in common in their configuration?' (2) 'How many electrons does each Group 1 element have in its outermost s orbital?' Part C — Reference Check (5 min): Pairs swap worksheets with an adjacent pair and cross-check against a teacher-projected answer key, marking any errors with a green pen (not crossing out — circling for correction).	tell us the pattern.' For Part C, project the answer key and say: 'Green pen only. Circling, not crossing out. Every error is information — it tells you where your mental model needs updating. That is the spirit of scientific thinking.' Circulate during cross-checking and kneel to eye-level with any pair who has more than 3 errors; quietly ask: 'Which step in the Aufbau order tripped you up?'		
Explain Phase 📹 **VIDEO:** [Electron configurations for the first period] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electron Cloud Atomic Model (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Part A — Peer-Teaching Carousel (15 min): Each pair is assigned one element from a set of 10 'focus elements' (Ne, Na, Mg, Al, Si, P, S, Cl, Ar, Ca). They have 3 minutes to rehearse explaining: (a) the full configuration, (b) how many electrons are in each subshell, and (c) which group the element belongs to and why. Pairs then rotate in a carousel: each pair stands at a station (desk with the element name card) and explains their element to the visiting pair. Visiting pairs must ask at least one question. After 6 minutes, pairs rotate once more. Part B — Think-Pair-Share: Phenomenon Reconnection (10 min): The teacher projects the anchoring photograph of the Kenyan cooking hearth alongside sodium and potassium configurations side by side (Na: $1s^2 2s^2 2p^6 3s^1$ and K: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$). Students silently think for 2 minutes, then discuss with their partner for 3 minutes, then share with the class: 'Using these configurations, explain why sodium and potassium both produce warm-toned flame colours, but sodium gives yellow-orange and potassium gives violet.' Part C — Individual Written	For the Peer-Teaching Carousel, say: 'You have 3 minutes to become the class expert on your element. I will be listening — I want to hear the words subshell, electrons, and group used correctly.' During carousel rotations, position yourself at the Ca or Cl station — the two hardest elements — and listen without intervening unless a factual error goes unchallenged. If an error persists, whisper to the explaining pair: 'Check your p subshell count for that period.' For the Think-Pair-Share, project the two configurations and the hearth photograph simultaneously. Say: 'Look at sodium — $3s^1$. Look at potassium — $4s^1$. One electron in the outermost s orbital in both cases. Now think: what is different? Think silently for 2 minutes — I want your best thinking before you speak.' Set a visible 2-minute timer. After pair discussion, cold-call 4 pairs in sequence, building on each response: 'Pair 1 — what is the same? … Pair 2 — what is different? … Pair 3 — how does that difference connect to colour? … Pair 4 — what would Kenyan ugali cooks need to know about	Peer-Teaching Carousel — When students must explain a concept to peers who will ask questions, they process the material at a deeper level than passive review (elaborative interrogation). The carousel format ensures every learner practises both explaining and questioning, distributing the cognitive load across the class. The subsequent Think-Pair-Share uses the anchoring phenomenon as the explanatory target, so all abstract notation ($1s^2 2s^2 2p^6 3s^1$) is immediately cashed out in terms of observable reality (yellow flame over ugali). The four required vocabulary terms in Part C scaffold written scientific explanation without over-constraining student voice.	Observable indicators: (1) During the carousel, each pair uses the words 'subshell,' 'electrons,' and 'group' correctly at least once (teacher tally). (2) In the Think-Pair-Share whole-class share, at least 3 of the 4 cold-called pairs can articulate that both Na and K have one outermost s electron but in different principal energy levels, leading to different photon energies and colours. (3) Individual written explanations in Part C contain all four required terms used in a chemically accurate sentence. Teacher collects 6 randomly selected books at the end of Phase 3 to review before Lesson 9.

	Explanation (5 min): Each learner writes 3–4 sentences in their exercise book answering the Think-Pair-Share question independently, using the terms: outermost electron, energy level, photon, wavelength.	this if they wanted to identify a mineral by flame colour?' For Part C, say: 'Close your eyes for 5 seconds and picture the yellow flame at the Rift Valley homestead. Now open your eyes and explain it in writing using the four terms on the board.' Allow 5 minutes of silent writing.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Electron configurations for the first period Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each pair receives two sticky notes. On the first (green): they write one 'Key Insight' — something they now understand about electron configuration that they did not understand at the start of the lesson. On the second (orange): they write one 'Lingering Question' — something that the lesson raised but did not fully resolve. Students walk to the DQB and post their notes in the designated columns. The teacher reads 4–5 green notes aloud and 3–4 orange notes aloud. The class then votes (thumbs up) on which lingering question they most want answered in Lesson 9. The teacher writes the winning question in a special 'Next Lesson Target' box on the DQB.	Distribute sticky notes and say: 'The green note is evidence that your thinking has changed. The orange note is proof that you are still curious — both are signs of a good scientist.' Allow 3 minutes for writing. As students post notes, stand at the DQB and read a selection aloud with commentary: 'This green note says — same outer electron, same group, same type of behaviour. That is exactly the insight chemists use to organise the entire periodic table. Hold onto that.' For orange notes: 'This pair asks — if sodium has one 3s electron and potassium has one 4s electron, does the extra energy level mean potassium's electron is easier to remove? That is a superb question — it connects directly to what we will explore in Lesson 9 on periodic trends.' Circle the winning question with a red marker and say: 'This is our target for our final evidence lesson. Come ready.'	Metacognitive DQB Protocol — The two-colour sticky-note system externalises both understanding and uncertainty simultaneously. By making 'not yet knowing' visible and valued (orange notes are posted publicly without shame), the teacher models the scientific norm that productive questions are as important as answers. Reading selected notes aloud and connecting them to the storyline reinforces that the DQB is a living document tracking collective sense-making — not a display board.	Observable indicators: (1) Every pair posts at least one green and one orange note — 100% participation. (2) Green notes reference at least one of the following: outermost orbital, group pattern, energy level difference, or flame colour connection. (3) Orange notes reveal whether students are ready for Lesson 9 — teacher scans for questions about ionisation energy, periodic trends, or orbital shapes, which confirm readiness for the next phase of the storyline.
Model Building Phase 📹 **VIDEO:** Electron configurations for the first period Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Students return to the cumulative atomic model diagram they have been building since Lesson 1. Today's task: (a) Annotate the sodium atom diagram with its full s and p configuration ($1s^2 2s^2 2p^6 3s^1$), labelling each subshell on the diagram; (b) draw a second diagram for potassium ($1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$) beside it, to	Say: 'Open your model-building booklet to your sodium diagram from Lesson 4. Today we make it complete. You now have the full language — s and p notation — to label every electron precisely.' Project a partially completed model on the board (with the $1s^2$ and $2s^2$ labels filled in but $2p^6$ and $3s^1$ left blank) and say: 'Your	Cumulative Model Annotation — Adding full s and p notation to a diagram students have been building since Lesson 1 creates a visible record of growing explanatory power. Placing sodium and potassium side by side on the same page makes the structural similarity (both end in ns^1) and the structural difference (n=3 vs n=4)	Observable indicators: (1) Sodium diagram is correctly labelled with all four subshells ($1s^2$, $2s^2$, $2p^6$, $3s^1$) and the outermost-electron arrow points to $3s^1$. (2) Potassium diagram correctly shows four principal energy levels with the outermost arrow at $4s^1$. (3) The written sentence correctly attributes the colour difference to a

for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the same scale; (c) add a colour-coded arrow on each diagram showing the outermost electron's position, and label it 'the electron responsible for the flame colour'; (d) beneath both diagrams, write one sentence connecting the electron arrangement to the anchoring phenomenon, using the sentence stem: 'Sodium produces yellow-orange flame because … whereas potassium produces violet flame because …'	diagram should look something like this — but yours will be fully labelled by the time you leave today.' After 6 minutes of individual work, say: 'Hold up your potassium diagram. Partner — check: does it have four principal energy levels? Does the outermost electron sit in the 4s subshell? If your partner's answer is no, help them fix it now.' Allow 2 minutes of peer correction. With 2 minutes remaining, cold-call 2 students to read their sentence stems aloud. Affirm: 'Notice — you just used electron configuration to explain a real fire at a Kenyan homestead. That is chemistry doing exactly what it should do.' Close with: 'In Lesson 9 — our final evidence lesson — you will use everything in this model to explain periodic trends and complete your full scientific explanation of the phenomenon. The colours of fire are almost fully explained. Come ready to finish the story.'	immediately comparable. The sentence stem forces students to translate notation into causal language, bridging the gap between symbolic chemistry and observable phenomena — the core scientific practice of constructing explanations.	difference in energy level (principal quantum number) of the outermost electron. Teacher collects model booklets from 5 students at the door as an exit-ticket check before Lesson 9.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the Peer-Teaching Carousel, which elements caused the most confusion — were they the same elements flagged in the Predict phase? If Ca and Cl were still problematic, plan a 5-minute 'hardest element' warm-up at the start of Lesson 9. (2) How many orange DQB notes mentioned ionisation energy or ease of electron removal? A high count signals the class is developmentally ready for periodic trends in Lesson 9 without needing additional scaffolding. (3) Did any learner with colour-vision deficiency struggle during the colour-coding activity? If so, introduce shape-coded symbols (circle for s-outer, triangle for p-outer) as an accessible alternative for future lessons. (4) Were the individual written explanations in Phase 3 (Part C) causally complete — did students connect energy level difference to photon energy to colour? If fewer than 70% of the 6 sampled books achieved this, begin Lesson 9 with a whole-class model sentence before moving to periodic trends. (5) Consider whether the Kenyan cooking-hearth context felt authentic and student-generated in discussions, or whether it required heavy teacher prompting — authentic student reference to the phenomenon (unprompted) is the strongest indicator that the storyline has taken hold.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed the complete s and p electron configuration pattern for all 20 elements (H–Ca) on their colour-coded Configuration Table, noting that all Group 1 elements (H, Li, Na, K) share a lone outermost ns^1 electron, and that Group 2 elements (Be, Mg, Ca) share ns^2, while halogens (F, Cl) share ns^2np^5.

What did I learn?	Students learned that the full s and p electron configuration for any element from H to Ca can be written systematically using the Aufbau order; that elements in the same group have identical outermost subshell types and electron counts; and that sodium ($3s^1$) and potassium ($4s^1$) differ only in the principal energy level of their lone outermost electron — which is why they produce similar but distinctly coloured flames.
How does this explain the phenomenon?	Students explained that the yellow-orange flame of sodium and the violet flame of potassium at the Kenyan cooking hearth arise because both elements have one electron in an outermost s orbital, but sodium's electron is in the third energy level and potassium's is in the fourth — meaning each releases a photon of a different energy (and therefore a different colour of visible light) when the excited electron falls back to its ground state.

LESSON 9 (80 minutes): Model Building, DQB Resolution, and Unit Consolidation: Answering the Driving Question	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all learning from Sub-Strand 1.2 by constructing a final cumulative atomic model, formally answering the driving question about flame colours and electron arrangement, and closing the Driving Question Board.
Knowledge	The learner describes the structure of the atom using Dalton, Rutherford, and Bohr models progressing to the modern s/p orbital model; explains isotopes and relative atomic mass; and writes electron configurations for the first 20 elements using s and p notation.
Skills	The learner constructs and presents an annotated cumulative model comparing atomic models across history; writes and justifies full electron configurations for sodium and potassium; evaluates which driving questions have been answered and identifies those remaining open for Sub-Strand 1.3.
Attitudes	The learner develops interest in the study of the structure of the atom and demonstrates integrity and social justice by contributing equitably to group model-building and presenting findings with confidence.
Key Inquiry Question	How are electrons arranged in an atom?
Purpose in Storyline	This is the final lesson of the sub-strand. Students bring together every strand of evidence gathered across Lessons 1–8 — atomic models, atomic number, mass number, isotopes, relative atomic mass, energy levels, orbitals, and s/p electron configurations — into one coherent cumulative model. They use that model to fully and formally answer the driving question: the different flame colours produced at the Kenyan cooking hearth (yellow for sodium, violet for potassium) arise because the outermost electrons of each element occupy orbitals at different energy levels, release different quanta of energy as photons of different colours when they fall back to ground state, and this is a direct fingerprint of the atom's electron arrangement. The DQB is formally closed and remaining questions are bridged to Sub-Strand 1.3: The Periodic Table.
Safety Notes	No new hazardous activities in this lesson. If students reference flame test results from earlier lessons, remind them that Bunsen burners must always be lit with striker only, loose clothing must be tied back, and chemical residues must be disposed of according to school lab protocol. Handle any printed or digital materials responsibly.

B. LESSON OVERVIEW

In this 80-minute final lesson, students work in established groups to build a cumulative poster or digital diagram — the 'Atomic Model Journey' — that traces atomic theory from Dalton's solid sphere through Rutherford's nuclear model and Bohr's shell model to the modern s/p orbital model. Each model is annotated with what it explained, what it could not explain, and the key evidence that forced scientists to revise it. Students add panels for isotopes, relative atomic mass calculation, and full s/p electron configurations for sodium (Na) and potassium (K). Groups then deliver 3-minute gallery-walk presentations, after which the whole class reunites to compose a collective written answer to the driving question on the board. Finally, the teacher leads a structured DQB closure: students vote on which wonder questions are now fully answered, which are partially answered, and which are passed forward to Sub-Strand 1.3. The lesson ends with individual exit slips that serve as the summative formative checkpoint for the sub-strand.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** The history of atomic chemistry Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown the anchoring phenomenon description one final time — the yellow flame from spilled salt at the Rift Valley hearth and the violet flash from banana-peel ash — displayed on the board or projected on screen. Without notes, each student silently writes for 3 minutes in their exercise books: 'Using everything you now know, explain in your own words WHY sodium produces yellow light and potassium produces violet light when burned.' Students then share their draft explanation with their seat partner (Think-Pair) before the class hears three cold-called responses. The teacher does NOT correct yet — these draft explanations are kept visible (written on sticky notes or mini-whiteboards) and will be compared with the final class explanation at the end of the lesson, making the growth in understanding tangible.	Display the phenomenon image/text and say: 'We started this unit eight lessons ago staring at this fire. Before we build anything today, I want to know — right now, no notes — what do YOU think is happening inside that flame? Write your best explanation. You have 3 minutes. Go.' [Allow 3 minutes silent individual writing. Do NOT circulate — let students retrieve independently.] After 3 minutes: 'Share with your partner. 1 minute.' Then cold-call exactly 3 students: 'Wanjiku, read me your explanation.' / 'Otieno, what did your partner say that you hadn't thought of?' / 'Amina, in one sentence — what is the KEY reason for the colour difference?' [10-second wait time after each question before accepting answers.] Write key phrases from student responses on the board under the heading 'OUR BEST EXPLANATION — START OF LESSON 9'. Say: 'We will come back to these words at the end. Let's see how our model changes.'	Strategy: Retrieval Practice + Visible Thinking. By forcing unaided recall before instruction begins, students activate and surface prior knowledge from Lessons 1–8. Displaying their initial explanations publicly creates a concrete before-and-after record that makes conceptual growth visible, motivating deeper engagement with model-building in subsequent phases.	Observable indicator: Every student produces a written explanation of at least 2–3 sentences within 3 minutes. Teacher scans the room to confirm all books are open and writing is occurring. Listen for presence of key terms in cold-called responses: 'energy levels', 'electrons', 'orbitals', 'photon/light', 'sodium', 'potassium'. Absence of these terms flags students who need targeted support during Phase 3.
Observe Phase 📹 **VIDEO:** The history of atomic chemistry Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12	Before building their final model, groups spend 10 minutes rotating through five 'Evidence Stations' set up around the classroom. Each station holds a one-page summary card (prepared by the teacher) representing a major evidence block from the unit: Station A — Dalton, Rutherford, Bohr model diagrams with key evidence for each; Station B — Atomic number, mass number, and proton/neutron/electron data table for Na and K; Station C — Isotope data for chlorine (Cl-35 and Cl-37)	Set up five stations before class. As groups rotate, circulate and prompt with targeted questions at each station. At Station A: 'Which model was the first to explain WHERE electrons are? What experiment forced that change?' [10-second wait.] At Station C: 'If I have two isotopes of carbon — Carbon-12 at 99% and Carbon-13 at 1% — would the RAM be closer to 12 or 13? Why?' [Cold-call one student.] At Station D: 'Point to the electron in sodium that gets excited in the flame. Which orbital	Strategy: Evidence Carousel with Annotation. Rotating through structured evidence stations mirrors the scientific practice of reviewing prior data before drawing conclusions (NGSS SEP 4: Analysing and Interpreting Data). The sticky-note annotation task requires students to actively connect each evidence piece back to the phenomenon rather than passively reading, reinforcing that all unit content is in service of explaining the Kenyan hearth flame colours.	Observable indicator: Each group's sticky notes at each station use at least one phenomenon-linked phrase ('…because sodium's 3s electron…' / '…this is why the colour changes…'). Teacher collects 2–3 sticky notes from Station D after rotation and reads them aloud (anonymised) to the whole class as a quick comprehension check before model building begins. Misconnections identified here are addressed immediately with a 60-second whole-class clarification.

🔍 Search ARES for similar readings	and the RAM calculation worked example; Station D — Energy level and orbital filling diagram ($1s^2 2s^2 2p^6 3s^1$ for Na; $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ for K); Station E — Flame test results table (colours for Na, K, Cu, Li, Ca) connecting back to the phenomenon. Groups spend 2 minutes at each station, adding sticky-note annotations: 'This connects to the flame colour because…' or 'This was the KEY evidence that changed the model because…' This rapid structured review ensures all groups have equal access to the full body of evidence before model construction.	is it in?' [Cold-call one student: 'Kamau, show me on the diagram.'] At Station E: 'Why does copper give a different colour from sodium? What does that tell us about copper's electron arrangement?' Give a 1-minute warning before each rotation. After all rotations: 'Return to your group seats. You now have ALL the evidence. It's time to build your final model.'		
Explain Phase 📹 **VIDEO:** The history of atomic chemistry Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups (4–5 students) spend 25 minutes constructing their 'Atomic Model Journey' poster or digital slide. The model must include five labelled sections: (1) Timeline of atomic models — Dalton → Rutherford → Bohr → Modern orbital model, with one key piece of evidence that forced each revision; (2) Annotated diagrams of sodium (Na) and potassium (K) atoms showing their full s/p electron configurations (Na: $1s^2 2s^2 2p^6 3s^1$; K: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$) and indicating the outermost (valence) electron; (3) Isotope panel — definition, example (Na-23 only stable isotope; note Cl-35/Cl-37), and RAM formula; (4) Flame explanation panel — a diagram showing an electron in the 3s orbital of Na absorbing heat energy, jumping to an excited state, then falling back and releasing a yellow photon ($\lambda \approx 589$ nm); an equivalent diagram for K's 4s electron releasing a violet	During construction (25 min): Circulate continuously. At each group, spend 90 seconds asking probing questions before moving on. Use these specific prompts: 'Show me on your diagram — which electron in potassium is the one that gets excited? How do you know?' / 'Your RAM panel says the RAM of chlorine is 35.5. Walk me through the calculation.' / 'Your flame diagram shows the electron going up — what form of energy does it absorb, and what form does it release? Are those the same?' [10-second wait after each.] If a group is stuck on the electron configuration of K, scaffold: 'Fill the orbitals in order: 1s, then 2s, then 2p — how many electrons fit in each? Keep going until you've placed all 19.' During Gallery Walk (12 min): Each group presents for 3 minutes. Teacher times strictly. After each presentation, cold-call one audience member: 'Akinyi, what was one connection their model	Strategy: Cumulative Model Construction + Gallery Walk (NGSS SEP 2: Developing and Using Models; SEP 6: Constructing Explanations). Building a single poster that integrates ALL unit content requires students to synthesise rather than merely recall. The Gallery Walk shifts the audience role to active listeners who must evaluate another group's model against their own, deepening metacognitive awareness. The teacher-facilitated construction of one shared class answer to the driving question on the board provides a whole-class consolidation anchor that ensures no student leaves with a fragmented explanation.	Observable indicator — Model completeness: Teacher uses a rapid 5-point checklist during gallery walk (1 point each): atomic model timeline present; Na and K electron configurations correct; isotope panel includes RAM calculation; flame diagram shows excited state and photon emission; driving question answered in writing. Groups scoring 3 or below are flagged for targeted support during Phase 5. Observable indicator — Presentation quality: Presenters use terms 'orbital', 'energy level', 'excited state', 'photon', and 'valence electron' correctly at least once each during their 3 minutes.

	photon; (5) Driving question answer — one or two written sentences directly answering 'Why do different substances produce different colours in a flame?' Groups then conduct a 3-minute Gallery Walk presentation to the rest of the class, with one student presenting while the others listen and use a peer-feedback card ('One thing they explained clearly' / 'One question I still have').	made that your group made differently?' After all groups: 'Let's write ONE class answer to the driving question on the board together. Who will give me the first sentence?' [Cold-call. Build the sentence collaboratively. Confirm it references: valence electrons, energy levels/orbitals, excited state, photon emission, different energies = different colours.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** The history of atomic chemistry Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electron Cloud Atomic Model (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher reveals the original DQB from Lesson 1 — the full collection of wonder questions students posted at the start of the unit (e.g., 'Why does salt make fire yellow?', 'Are all atoms of the same element identical?', 'What are electrons actually doing inside an atom?', 'Why do atoms have different masses?'). Each question card is read aloud. Students individually (no group discussion) assign each question to one of three categories using coloured dots or a show of hands: GREEN — fully answered in this unit; YELLOW — partially answered, more to learn; RED — not yet addressed, passes to a future strand. After voting, the teacher leads a brief discussion on 2–3 contested questions. The teacher then explicitly bridges RED and YELLOW questions to Sub-Strand 1.3: 'You asked WHY elements behave differently in reactions — that is exactly what the Periodic Table in our next sub-strand will explain, using the very electron configurations you've just drawn.'	Unpin or project the original DQB. Say: 'These are the questions YOU asked in Lesson 1 when you first saw the burning hearth. Let's see how far we've come.' Read each question card aloud with deliberate slowness. For each: 'GREEN if fully answered — show me.' [Count hands.] 'YELLOW if partially answered.' 'RED if we haven't touched it yet.' After voting on all cards: Select one GREEN question (e.g., 'Why does salt make fire yellow?') and cold-call: 'Mwangi, answer that question right now — fully — as if explaining to your grandmother in Kericho who has never studied chemistry. Go.' [15-second wait. Accept answer. Prompt the class: 'Did he answer it fully? What would you add?'] Select one RED question and say: 'This question is not unanswered — it is waiting for you. Write it in the first page of your new Sub-Strand 1.3 exercise book section as your personal driving question.' Formally close the DQB: 'This board is done. The fire mystery is solved. But bigger mysteries are waiting.'	Strategy: Structured DQB Closure with Forward Bridge (NGSS SEP 1: Asking Questions; SEP 8: Obtaining, Evaluating, and Communicating Information). Returning to the original questions from Lesson 1 creates a powerful narrative arc: students see their own intellectual journey made concrete. The three-category voting (fully answered / partially / not yet) builds metacognitive awareness about the limits of current knowledge — a core scientific practice. The explicit bridge to 1.3 prevents the DQB from feeling like a dead end and sustains curiosity.	Observable indicator: Every student participates in the voting (show of hands or dot stickers — teacher scans for non-participants and prompts them by name). Quality indicator: When cold-called to answer the GREEN question, the student's explanation references valence electrons, excitation, photon emission, and the phenomenon — all four elements. Absence of any one element is noted and addressed in the teacher reflection.
Model Building Phase	In the final 10 minutes, each student independently completes a structured Exit Slip (written on a	Distribute exit slip sheets (or direct students to a specific page in their books). Say: 'This is YOUR work.	Strategy: Individual Summative Exit Slip as Metacognitive Consolidation (NGSS SEP 6:	Observable indicator — Task 1: K configuration is written correctly as $1s^22s^22p^63s^23p^64s^1$ (19 electrons

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D. TEACHER REFLECTION

After collecting exit slips, evaluate the following: (1) What percentage of students correctly wrote the full s/p configuration for potassium ($1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$)? If fewer than 70% were correct, plan a 10-minute orbital filling drill at the start of the first Sub-Strand 1.3 lesson. (2) Did students' Task 2 flame explanations include all four required elements (4s electron, excited state, photon, energy/colour)? Which element was most commonly missing? This reveals which phase of the excitation-emission model remains least consolidated. (3) Review the 'Questions for Sub-Strand 1.3' collected in Task 4 — do these cluster around ion formation, periodic trends, or element families? Use these to personalise the opening phenomenon for Sub-Strand 1.3. (4) Reflect on equity of participation during the Gallery Walk: did all group members present, or did one or two students dominate? Consider restructuring group roles for 1.3. (5) Did the DQB closure feel rushed? If the Forward Bridge to 1.3 was cut short, plan to begin the next sub-strand with an explicit 5-minute DQB handover activity. (6) Note which groups produced the strongest cumulative models and consider displaying these in the lab as reference artefacts for Sub-Strand 1.3 electron configuration work.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed (via evidence station cards) flame test colour data for sodium, potassium, copper, lithium, and calcium; Rutherford gold-foil scattering diagrams; orbital filling diagrams for Na and K; and isotope abundance tables for chlorine — all

	gathered across Lessons 1–8 and reviewed in this lesson's evidence carousel.
What did I learn?	Students learned that each atomic model (Dalton → Rutherford → Bohr → modern s/p orbital model) was driven to revision by specific experimental evidence; that sodium's lone $3s^1$ valence electron and potassium's lone $4s^1$ valence electron occupy orbitals at different energy levels, so when excited by flame heat they release photons of different energies corresponding to different colours (yellow at ~589 nm for Na; violet for K); that isotopes of the same element differ in neutron number and thus mass number, and that relative atomic mass is a weighted average of isotopic masses; and that the electron configurations of the first 20 elements can be written systematically using s and p notation.
How does this explain the phenomenon?	Students explained the anchoring phenomenon — the yellow-orange glow when salt is spilled into the Rift Valley cooking fire and the violet flash from banana-peel ash — by constructing cumulative annotated models showing that sodium's outermost electron ($3s^1$) and potassium's outermost electron ($4s^1$) absorb thermal energy from the flame, jump to higher-energy excited states, and release that energy as photons of specific wavelengths when returning to ground state; because sodium's and potassium's valence electrons occupy orbitals at different energy levels, the energy — and therefore the colour — of the emitted photon is different for each element, making flame colour a direct fingerprint of atomic electron arrangement.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.